

06 COORDINATION

Coordination is the process of linking up processes in the body of an organism such that various processes occur in an organized manner.

KEY TERMS

- (i) **Irritability/sensitivity**; is the ability of an organism to detect and respond to changes in the environment. These changes are known as **stimuli**.
- (ii) **Stimulus**; is and change in the environment that causes change in the organism's pattern of behaviour. The stimulus is either internal or external.
- (iii) **Environment**; is the immediate surrounding of a cell or an organism. It's either internal or external.
 - (a) **Internal environment** is the immediate surrounding of cells-that's, body fluids such as blood, tissue fluid or lymph that bathe the cells of an organism.
 - (b) **External environment** is where an organism lives like air (atmospheric air) for terrestrial/land organisms and water for aquatic organisms.
- (iv) **Receptor**; is a cell, tissue or organ that detects a change in the environment. Ideally, receptor is a modified nerve cells that receive stimuli from the environment. It usually converts stimulus energy into electrical energy(message) in form of a nerve impulse by a process known as **transduction**. Hence receptors are known as **biotransducers**.
- (v) **Effector**; is a cell, tissue, organ or gland that produces a response to a particular stimulus. Ideally, effectors are modified cells that are capable of producing a response.
- (vi) **Hormone**; is a chemical message transmitted in an organism.
- (vii) **Transduction**; is a process by which stimuli are converted into electrochemical messages called nerve impulses.

COORDINATION IN PLANTS

Plants are coordinated organisms just like animals. However, unlike animals which rely on both neural(nervous) and hormonal(chemical) coordination, plants lack a rapidly responding nervous system. Therefore, they mainly exhibit chemical coordination which is slow and involves growth responses responsible for plant growth. Plant growth is regulated by chemical substances such as hormones with the following key characteristics;

- ✓ They are required in minute quantities.
- ✓ Their actions are specific.
- ✓ They are short-live and hence, don't accumulate since are constantly broken down.

- ✓ Some don't move away from their site of synthesis. However, most of them are transported both via vascular tissues and from cell to cell. Therefore, communication in plants can occur between cell to cell and between plants and plants.

These chemical substances are commonly called **plant growth substances** or **plant growth regulators**. However, the term “**plant growth regulators**” usually refers to their **synthetic versions** and it also includes others like **phytochromes** such as phytochrome red(Pr) and phytochrome far red(Pfr). Plant growth substances are also called **hormones** (**plant hormones/phytohormones**) because they carry chemical information from place to place by mass flow, active transport or diffusion.

There are five (5) major plant growth substances (plant hormones) namely;

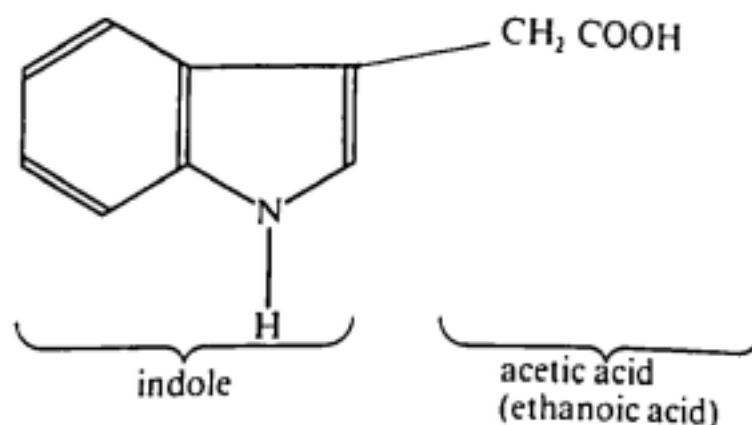
1. **Auxins**; responsible for cell enlargement and differentiation.
2. **Gibberellins**; for cell enlargement and differentiation.
3. **Cytokinins**; for cell division.
4. **Abscisic acid (ABA)**; for dormancy.
5. **Ethene/ethylene**; for ageing(senescence) and fruit ripening.

AUXINS

These are grouped under a class of compounds called **growth promoters**. They are growth stimulants whose small quantities have powerful effects on plant growth and development.

They **naturally** exist as **indole acetic acid (IAA)**.

Structure of IAA



Synthesis and distribution

They are mainly made in the shoot apex and transported via phloem tissue to roots. Their movement is polar—that's, occurs in one direction within phloem tissue and from cell to cell. They are inactivated or degraded by enzymes.

Key role: They control **phototropism**.

Effects

- Have an effect on cell division in combination with cytokinins.
- Stimulate cell differentiation.
- Stimulate cell elongation/expansion this is the sole effect of auxins.
- Inhibit growth of lateral buds (as they promote apical dominance)
- Initiate growth of adventitious roots.
- They stimulate the development of a callus; undifferentiated tissues; burying roots of stems.
- Promote parthenocarpy; development of fruits without fertilization.
- Retard abscission; fruit-fall.
- They initiate secondary growth in presence of other plant hormones.
- Initiate fruiting.
- Inhibits sprouting in potatoes.

Mode of action

Auxins stimulate secretion of hydrogen ions out of cell into the cell wall, increasing its acidity, providing the optimum low pH of activation of enzymes called **expansins** which then catalyze the breaking of bonds of cell wall polysaccharides like pectin and hemicellulose; thereby loosening the cell wall microfibrils; thus, the cell wall stretches more easily and expands due to osmotic intake of water in the cell as a result of low water potential that leads to high pressure potential thereby expanding the protoplast. They also stimulate laying down of new cell wall material. Likewise, auxins attach to receptors on the cell membrane of epidermal cells; causing them to change their gene activity leading to growth. Flavin like FAD which is a derivative of vitamin B₂ forms the receptor.

KEY TAKEAWAY

Auxins increase the plasticity of the cell wall by making it to stretch more easily thereby allowing cell expansion and growth.

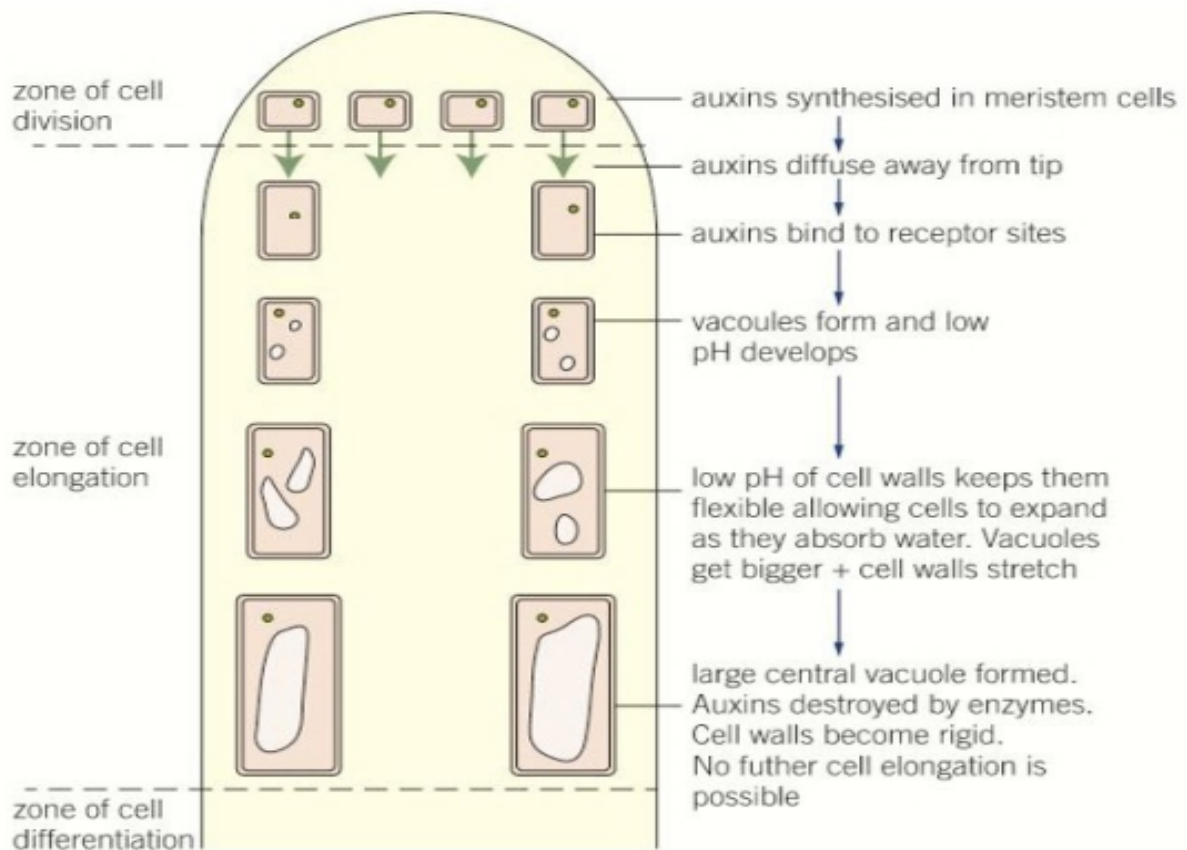
How?

Auxins bind to specific receptor sites in the plant cell membrane, causing efflux of hydrogen ions and a fall in pH of the cell wall to 5; the optimum pH for the expansin enzymes to keep the cell wall very plastic and flexible. On maturation of cells, auxins are destroyed, pH increases so expansin enzymes are become inactive; Thus, the cell wall becomes rigid and more fixed in shape and size so that cells no longer expand or grow.

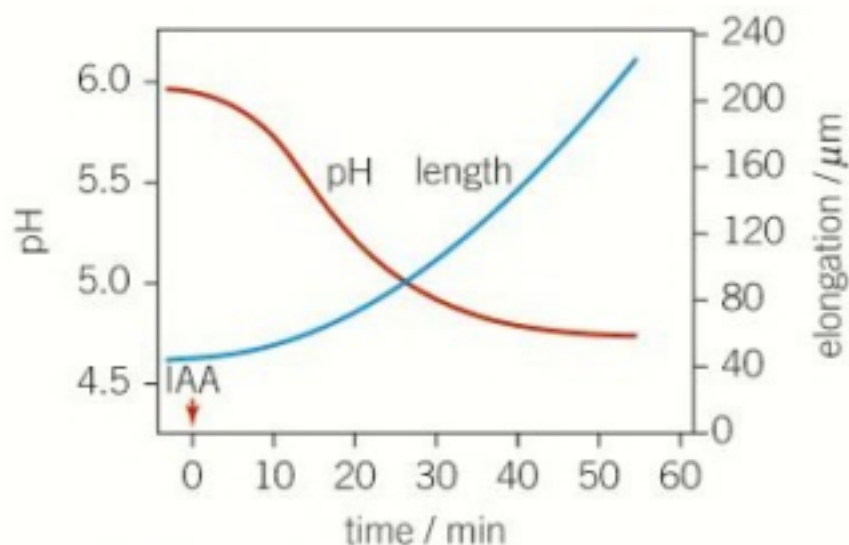
NB: Synthetic auxins include;

- ✓ Phenoxyacetic acid (2,4-D).
- ✓ Indole butyric acid (IBA)
- ✓ Indole propionic acid (IPA)

A diagrammatic illustration of the effect of auxins on apical shoot growth



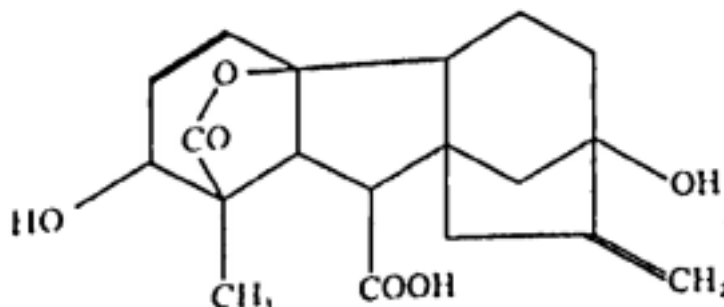
A graph showing the effect of an external application of auxin on pH levels in the cell walls and on shoot growth



GIBBERELLINS

These also belong to a class of compounds known as **growth promoters**. They are all terpenes with branches of weak acids, thus, terpenoids. Naturally, over 50 gibberellins occur with Gibberellic acid (GA_3) being the most common. They have a **gibbane skeleton**.

Structure of gibberellic acid



Synthesis and distribution

They are made in chloroplasts of young apical leaves, buds, seeds and root tips. Their movement is non-polar-that's, is bidirectional, occurring up and down in phloem and xylem tissues.

Key role: They break dormancy of certain seeds especially cereals.

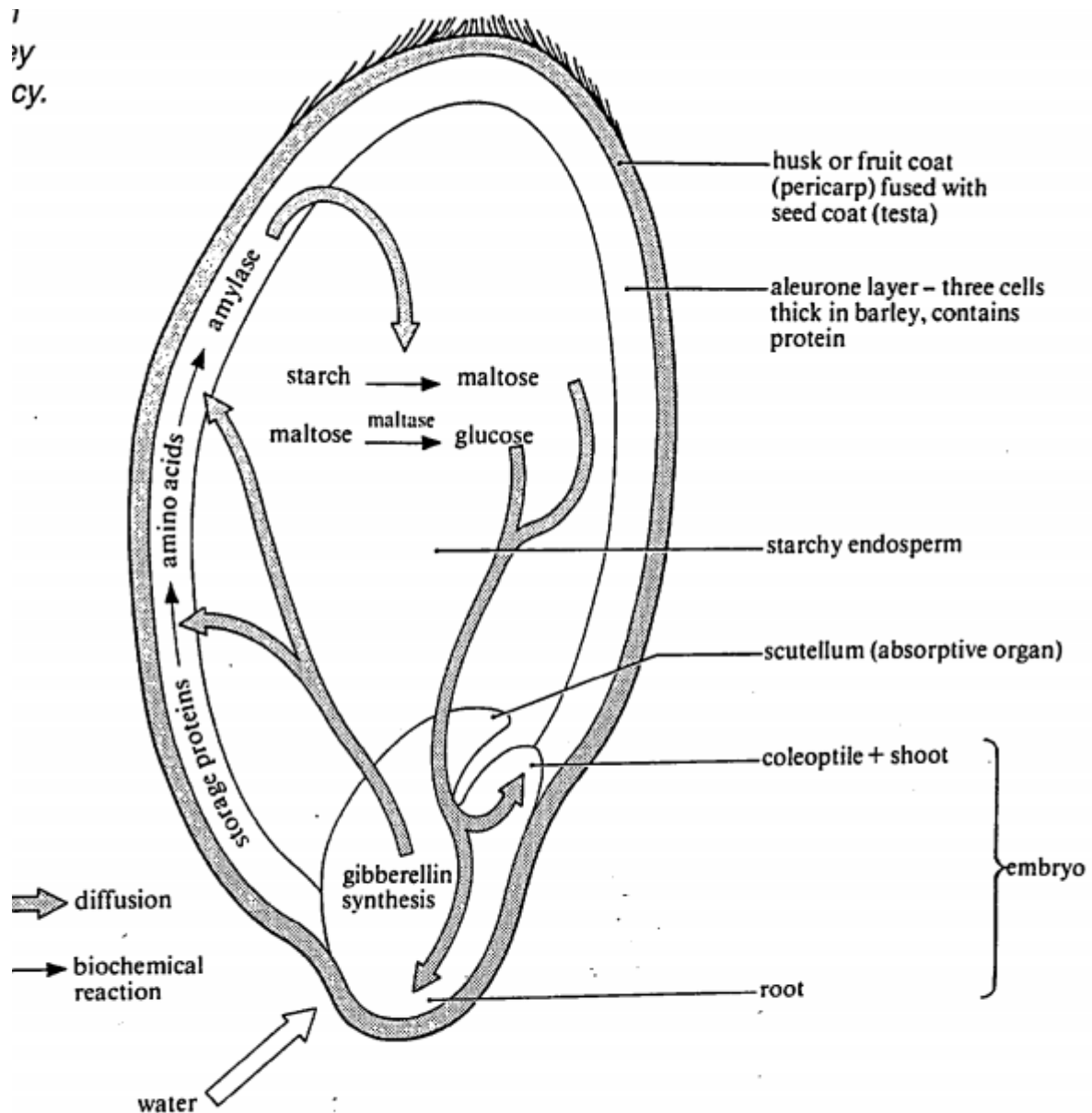
Effects

- Break seed dormancy.
- Promote parthenocarpy
- Break bud dormancy
- Control flowering in short day plants.
- Induce bolting in rosette plants by shortening the internodal length e.g. for the cabbages.
- Promote growth of side branches.

THE ROLE OF GIBBERELLINS IN MOBILIZING SEED RESERVES TO BREAK SEED DORMANCY

Imbibition of water hydrates and activates the embryo to start producing gibberellins; which are secreted and diffuse to the aleurone layer; where they stimulate the synthesis of enzymes like carbohydrases (e.g. alpha amylase), lipases and proteases from the stored proteins; where carbohydrases catalyze the hydrolysis of carbohydrates to simple sugars, lipases catalyze the hydrolysis of lipids to fatty acids and glycerol while proteases catalyze the breakdown of proteins to amino acids in the food stores (cotyledons in dicot seeds or endosperm in monocot seeds). Amino acids are re-organized to form growth hormones, more enzymes and structural proteins; sugars, fatty acids and glycerol are translocated to the growing centre, embryo where they are oxidized to provide energy in form of ATP for building materials used for active growth & development as well as breaking the seed coat by the embryo.

Illustration

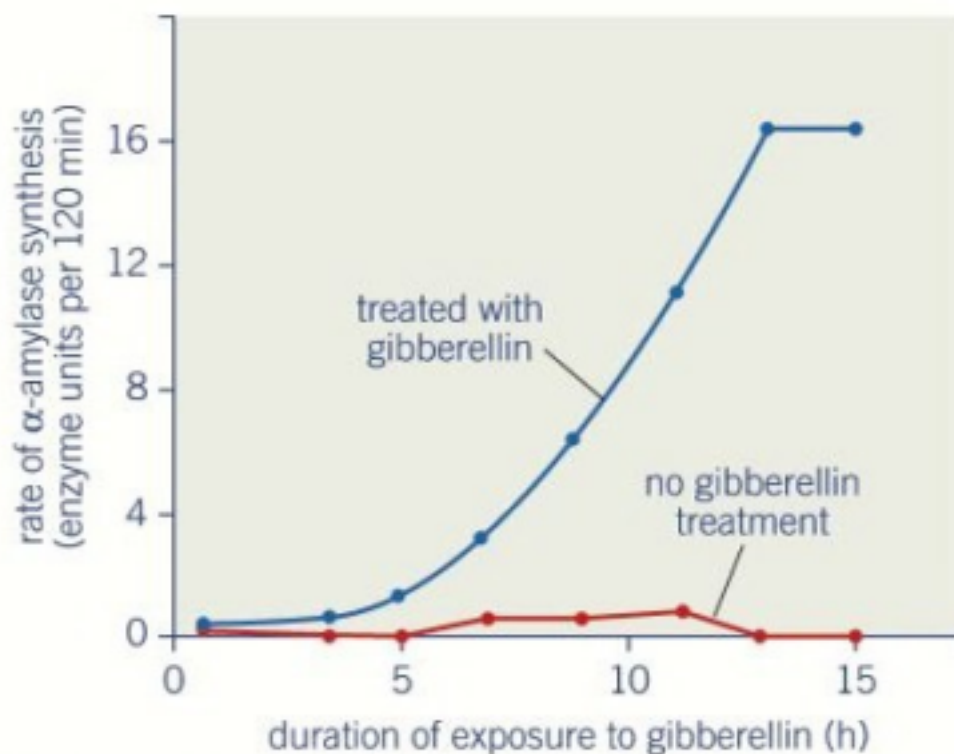


Mode of action

Gibberellins stimulate the synthesis of new proteins by switching on and off certain genes in cell differentiation. They actually switch on genes which code for amylase and proteases, the digestive enzymes required for seed germination.

NOTE: ABA is an antagonist to gibberellins thus, inhibits seed germination.

A graph showing the effect of gibberellins on the synthesis of amylase in isolated tissue from barley seeds

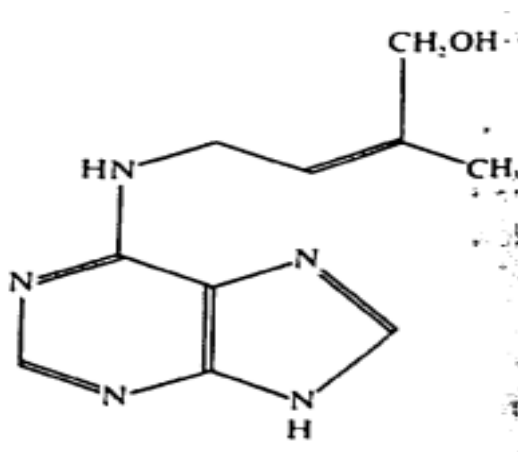


CYTOKININS

These are also grouped under a class of compounds called **growth promoters**. They are chemically similar to base adenine.

They naturally exist and **zeatin**. However, **kinetin** is a synthetic cytokinin.

Structure of zeatin



Synthesis and distribution

Cytokinins are made in roots and transported via xylem tissue to shoots; where they are exported from leaves via phloem. They are found in fruits and seeds where rapid cell division occurs.

Key role: They delay senescence by preventing disintegration of chlorophyll, maintaining protein synthesis and stimulating the breakdown of carbohydrates.

Effects

- Promote cell division(cytokinesis)
- Promote cell differentiation in developing roots, stems, leaves and flowers; in combination with auxins.

HINT: If the auxin-cytokinin combination is such that;

- ✓ Cytokinins (in big amounts; relatively) above auxins, then the shoot differentiates.
- ✓ Cytokinins less than auxins, cell differentiation in roots
- ✓ Cytokinins are equal to auxins, cell differentiation occurs in both roots & shoots.
- ✓ Intermediate cytokinins (cytokinins not too much) and very few auxins, cell division(mitosis) occurs not differentiation.

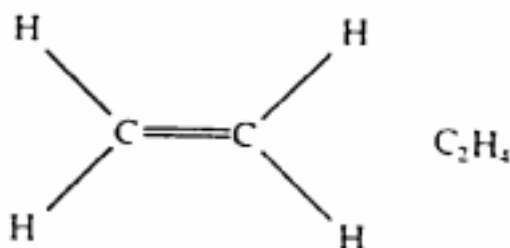
Mode of action

They participate in synthesis of nucleic acids and transfer RNA.

ETHENE/ETHYLENE

This is a gaseous hormone thus, its easy transportation to all plant organs.

Structure of ethane molecule



SYNTHESIS AND DISTRIBUTION

It's found in most organs. Although it is a gas, it doesn't move in intercellular spaces. It escapes from plant surface and the precursor of ethane is transported in xylem up to the shoot.

Key role: It speeds up ripening of citrus fruits.

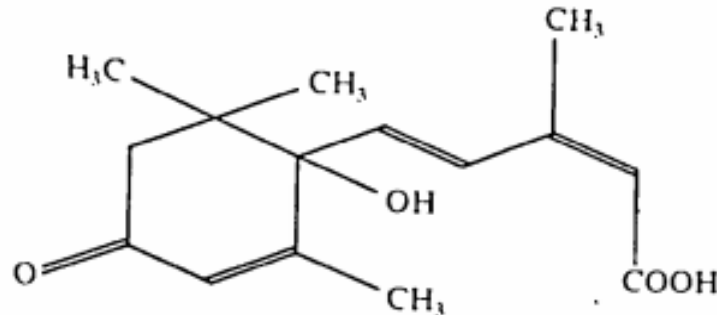
Effects

- Causes fruits ripening.
- Wound healing in plants
- Causes leaf-fall; abscission of leaves.
- Causes fruit-fall; abscission of fruits.
- It releases buds and seeds of certain plants from dormancy.
- **NB: Abscission** is the falling of fruits and leaves.

ABSCISIC ACID (ABA)

This compound is classified as a **growth inhibitor**. It has similar roles like those performed by lunularic acid in algae and liverworts. It is a terpenoid and is the only growth substance in its class.

Structure of ABA



Synthesis and distribution

It's made in leaves, stems, fruits and seeds and transported in phloem tissue to the root cap. It moves from the root cap by diffusion.

Key classical role: It induces dormancy.

Effects

- Inhibits division of vascular cambium (secondary growth).
- Abscission
- Closure of stomata in leaves during water storage.
- Promotes seed dormancy.

Mode of action: Is unknown.

THE ROLE OF PLANT GROWTH SUBSTANCES IN PLANT GROWTH AND DEVELOPMENT

Plant regulators control growth and development by affecting tropisms especially phototropism and geotropism as well as **phototropism**.

THE ROLE OF PLANT HORMONES IN TROPISMS

Tropism is the **growth movement** of a plant part in response to and directed by an external **unidirectional stimulus**. A tropic response is named according to the nature of the stimulus; whereby plant growth movement towards a stimulus is a **positive tropic response** while plant growth movement away from the stimulus is a **negative tropic response**.

Our scope here is limited to the discussion of only two tropic movements-that's, **phototropism** for light and **geotropism/gravitropism** for gravity.

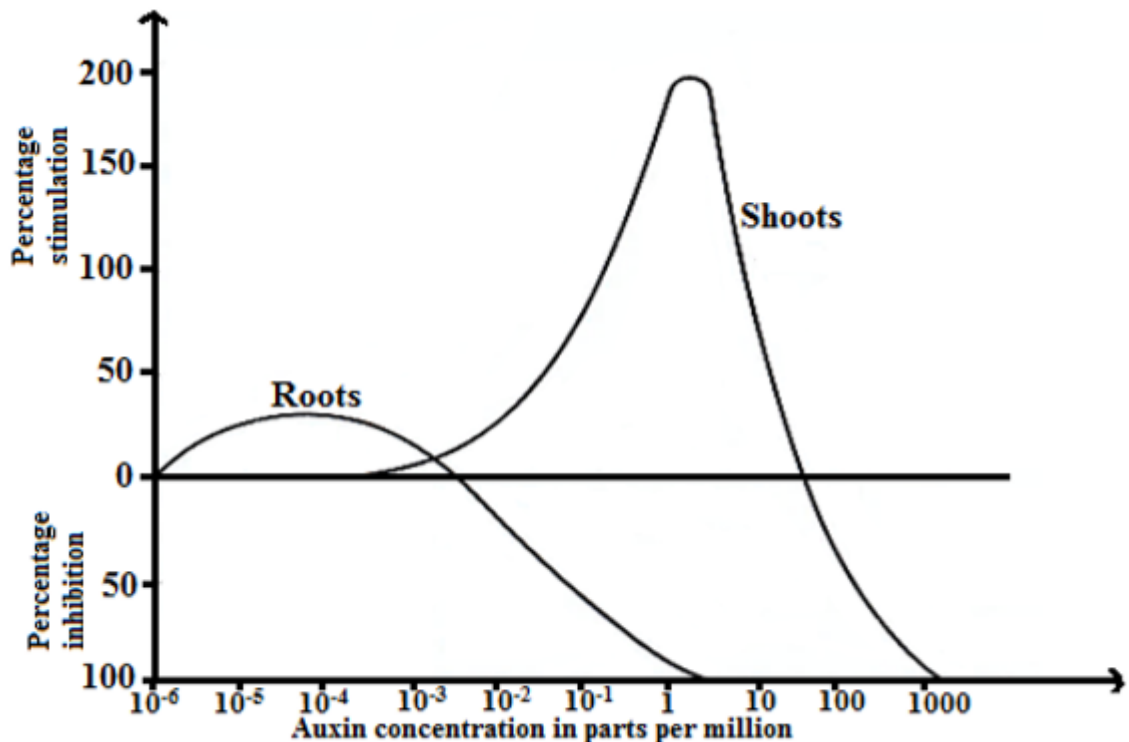
AUXINS AND PHOTOTROPISM

Phototropism is the plant growth movement towards or away from unidirectional light.

The role of auxins in phototropism

Consider the graph below;

Effects of concentration on auxins on cell elongation in roots and shoots in oat seedlings



In shoots, low auxin concentration inhibits growth, high auxin concentration stimulates growth and very high auxin concentration inhibits growth.

In roots, low auxin concentration stimulates growth, moderate auxin concentration decreases stimulation of growth and high auxin concentration inhibits growth.

Therefore, shoots and coleoptiles are positively phototropic and bending occurs in the region of cell elongation while many roots are negatively phototropic such as adventitious roots of climbers like ivy.

NB: Don't use the word "inhibit" in the region of stimulation, just use decreasing stimulation.

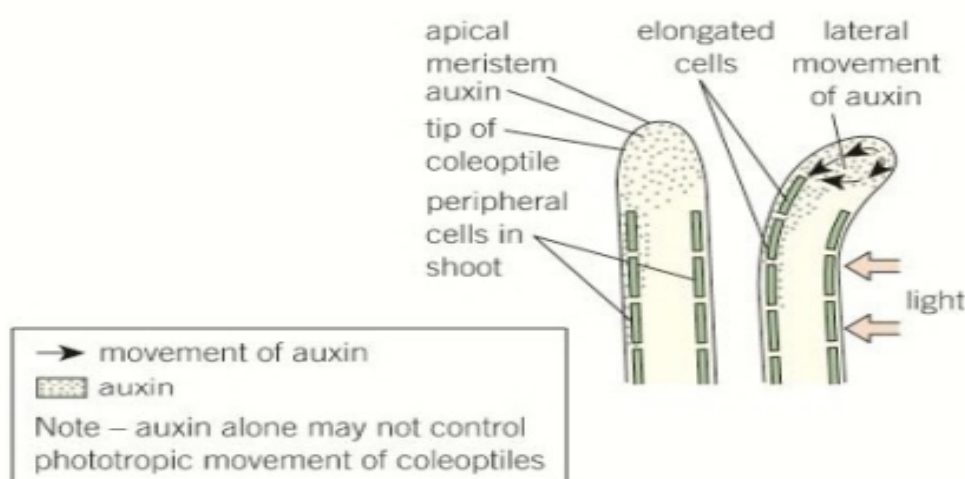
Explanation

In shoots, light is perceived by the tips, which secrete auxins; auxins diffuse down to the zone of cell elongation; where they stimulate or inhibit cell expansion/elongation and hence growth depending on their concentration. Whereby; in darkness/uniform illumination, auxins diffuse equally down on all sides, leading to stimulation of uniform

growth thus no bending. In non-uniform/uneven illumination, auxins diffuse away from light/illuminated side to dark side/shaded area/unilluminated side leading to greater/higher auxin concentration on dark/shaded/unilluminated side, which stimulates faster cell elongation and hence growth on the dark side than the illuminated side thus, curvature/bending towards the light source.

In roots, auxins diffuse away from the light side to the dark side leading to their higher concentration on the dark side where they inhibit growth and a consequently faster growth on the illuminated side and hence, bending away from light into the soil.

An illustration of a phototropic response in a shoot



NOTE:

- (i) In each case consider; the stimulus (light)→ receptor (flavin at the tip)→ transmission(diffusion))→ effector (zone of cell elongation))→ response (cell elongation/expansion).
- (ii) The light receptor is a Flavin like coenzyme FAD from vitamin B_2 which absorbs light from blue spectrum, leading to redistribution of auxins.
- (iii) Beyond optimum auxin concentration-that's, $1(10^0)$ for shoots and 10^{-5} for roots, further increase in auxin concentration inhibits growth.

ADVANTAGES/SIGNIFICANCE OF TROPIC RESPONSES OF SHOOTS AND ROOTS

For shoots

- Enables leaves to be exposed to light which is the source of energy for photosynthesis.
- Enables flowers to be exposed to pollinating agents for successful pollination and fertilization.
- Enables fruits to be exposed to dispersal agents for easy dispersal.

For roots

- Enables the exposed roots grow towards soil or equivalent suitable substrate for firm anchorage; absorption of enough water and mineral salts.

GEOTROPISM

This is the plant growth movement of a plant part towards or away from gravity. It is also known as **gravitropism**.

Plants are also sensitive to **gravity**. Hence, gravity controls **plant growth** by affecting **different responses** of the roots and shoots. Gravity always acts downwards, so plants always receive a unilateral gravitational stimulus. Shoots are usually negatively geotropic-that's, grow away from gravitational pull while many roots are positively geotropic-that's, grow towards the gravitational pull. Therefore, shoots grow up to the light while most roots grow down into the soil.

AUXINS AND GEOTROPISM

Auxins distribution is also due to gravity (direction of gravitational pull).

Roots are positively geotropic-that's, they grow in the direction of action of gravity.

Explanation

Gravity causes the auxins to move/migrate to the lower side of the root/radicle, where higher auxin concentration inhibits growth while lower auxin concentration stimulates faster growth by cell elongation/expansion on the lower side, leading to growth of the roots/radicles downwards.

The opposite is true for shoots whereby shoots and coleoptiles are **negatively geotropic**.

In shoots, the gravity receptors called statoliths are located in the zone of elongation but not at the tip. So, they sediment there in response to gravity; allowing auxins also to accumulate there, which stimulates growth leading to upward growth of the shoot or coleoptile.

NOTE: Auxins diffuse in response to the gravity (to the direction of gravitational pull).

GRAVITY SENSING MECHANISM

THE STARCH-STATOLITH HYPOTHESIS

Gravity is perceived by receptors called **statoliths**. In roots, the root cap has large starch grains in amyloplasts of cells called **statocytes**. The statocytes have the starch grains known as **statoliths**, which are the receptors for gravity (gravitational pull). The statoliths transmit the response as they sediment/move down in response to gravity. This sedimentation affects the distribution of growth substances especially auxins from the root apex. Decapitation of the root tip removes sensitivity to gravity due to loss of statoliths (receptors). However, in shoots, statoliths are located in the zone of elongation but not at the shoot tips as they sediment there in response to gravity.

HINT: If the seedling (root & shoot) is placed horizontally, auxins diffuse to the lower side of the shoot and root. In the shoot, high auxin concentration stimulates faster growth on the lower side than on the upper side; so, the shoots grow upwards. In the root, high auxin concentration inhibits growth on the lower side; so, the root grows downwards.

ADVANTAGES/SIGNIFICANCE OF GEOTROPISM TO PLANTS

Enables shoots of germinating seeds grow upwards through soil towards light for photosynthesis.

Roots are positively geotropic thus, roots penetrate soil for firm anchorage and absorption of water and mineral salts.

Rhizomes and runners are diageotropic which enables plants to colonize new areas of soil.

Dicotyledonous leaves are diageotropic, enabling the flat surface of the leaf to absorb maximum sunlight due to maximum exposure at right-angles to incident radiation.

Lateral roots are plagiogeotropic, allowing a large volume of soil to be exploited and the arrangement of roots to provide enough support similar to guy-ropes supporting a tent.

Branches are plagiogeotropic, allowing a larger volume of space to be occupied for maximum exploitation/absorption of light.

EFFECT OF ABA ON GEOTROPISM

ABA is transported from root tips to other tissues by lateral transport in response to gravity; thereby inhibiting growth there unlike IAA which don't undergo lateral transport. So, ABA inhibits geotropism.

MODERN HYPOTHESIS ON GEOTROPISM

We still have geotropism inhibited on removal of root tip. This is because the growth inhibitor, ABA is transported from the root cap to zone of elongation and inhibits growth there. This kind of transport also occurs together with ethane and gibberellins.

SUMMARY OF THE EFFECTS OF PLANT HORMONES

Hormone	Effects		Examples
Auxins	Promote cell elongation	Phototropic responses	Shoots bend towards light.
		Geotropic responses	Roots grow down into the soil.
	Stimulate cell division	Promote development of roots	Hormone in rooting powders used on stem cuttings.
		Stimulate cambial activity	Callus development at site of wounds.
		Stimulate development of fruits	Natural fruit setting is improved by spraying synthetic auxins. Fruit development by parthenocarpy especially in apples.
	Maintain cell wall structure	Inhibit leaf abscission	If supply of auxins from leaves exceeds that from stems.
		Inhibit fruit abscission	If supply of auxins from fruits exceeds that from stems.
	Inhibit growth in high concentration	Apical dominance	Lateral buds remain dormant under the influence of auxins in apical bud
		Disruption of growth	Synthetic auxins like 2,4-D and 2,4,5-trichlorophenoxyacetic acid are used as selective weed killers.
	Promote enlargement of cells		Stem growth

Gibberellins	Reverse genetic dwarfism	Dwarf varieties of peas, maize grow to normal size on gibberellin application
	Promote cell elongation	Increase length of internodes; blotting in cabbages turned to rosettes shape.
	Break dormancy of buds.	E.g. birch(Betula).
	Break seed dormancy	e.g. Ash, wheat, barley as a result of mobilizing food reserves that leads to germination.
	Stimulate fruit development	E.g. In cheny, peach and by pathenocarpny in apples and pear.
	Remove the need for cold treatment in vernalization	E.g. In carrots.
	Affect flowering	Flowering is promoted in some long day plants thus substitute for red light. Inhibit flowering in short day plants; change the sex of flower in monoecious plants.
Cytokinins	Promote cell division	Increase growth in sunflower(Hellanthus).
	Delay leaf senescence	Maintain leaf for some time after being detached.
	Stimulate bud development	On cuttings e.g. African violet (saint paulia) and mosses.

	Break dormancy	Of seeds and buds.
Absciscic acid(ABA)	Inhibits growth.	Retards growth in most plants especially in drought and physiological stress
	Promotes senescence	Promotes ageing.
	Induces dormancy	Of seeds and buds e.g. in birch, sycamore(Acer).
	Causes stomatal closure.	Under water stress(wilting).
	Promotes abscission	Of leaves and fruits.
Ethene	Ripens fruits	Of most citrus fruits like bananas which are stored unripe, transported and then apply ethene to ripen them.
	Induces flowering	In pineapples. Affects sex of flowers in dioecious plants e.g. in cucumber it changes male to female flowers.
	Inhibits stem elongation	But allows transverse growth thus, swollen stems formed.
	Promotes senescence	Of leaves and fruits(ripening)
	Promotes abscission	Of leaves and fruits.
	Breaks dormancy	Of buds in some plants.

COMMERCIAL APPLICATIONS OF PLANT GROWTH SUBSTANCES

There are several applications of plant hormones which include the following;

1. Control of ripening

Ethene is used as **ripeners** of climacteric fruits. **Climacteric fruits** are the ones that continue to ripen after being harvested. These include bananas, tomatoes, mangoes and avocados. Their ripening is linked to a peak of ethene production which triggers a series of chemical reactions involving a great increase in the rate of respiration, unlike that of non-climacteric fruits. **Non-climacteric fruits** don't produce large amounts of ethene and don't ripen much after harvesting or picking. For example; oranges, strawberries and watermelon.

Therefore, ethene is commercially used to produce perfectly ripe climacteric fruits for greengrocers and supermarkets.

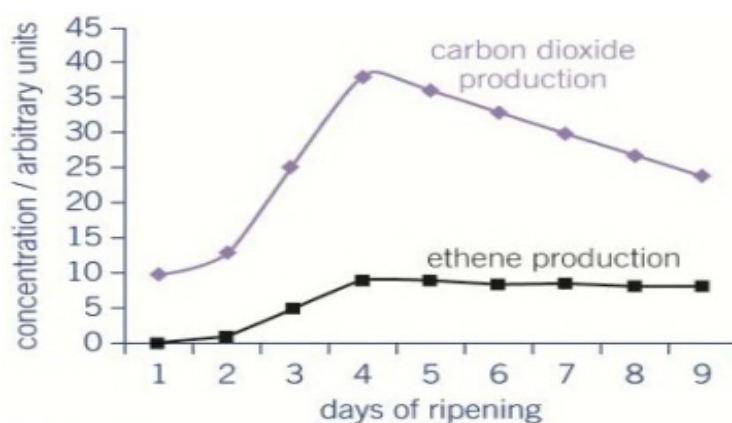
How?

These fruits are harvested when fully formed but long before they are ripe (harvested when mature but unripe) then are cooled, stored and transported to their destinations (even worldwide). This is because unripe fruits are hard and much less easily damaged during transportation than ripe ones. When they reach their destinations and needed for sale, they are exposed to ethene gas under controlled conditions especially of temperature. This ensures that each batch of fruits ripens at the same rate and all fruits are at the same stage to be put on the shelves for sale to the public. Therefore, this careful control of fruit ripening prevents a lot of fruit wastage during transportation and increases the time available for them to be sold (shelf-life).

EFFECT OF ETHENE ON CLIMACTERIC FRUITS

Consider 2 bunches of green bananas. 1 placed in a polythene bag with a single ripe banana finger and the other is put in a second polythene bag without a ripe banana finger. The bunch with a ripe banana finger will ripen faster than that without a ripe banana finger at the same temperature. This is because ethene being a gas, it diffuses from the ripe banana finger to green ones and stimulates their rapid ripening.

A GRAPH SHOWING THE EFFECT OF ETHENE ON RESPIRATORY RATE OF A RIPENING FRUIT



2. **Control of rooting**

Auxins are used as rooting powders to stimulate formation of roots from stem/vine cuttings for faster propagation of new plants from plant cuttings.

3. **Control of weeds (weeding)**

Many staple or commercial food crops are mainly narrow-leaved monocotyledonous plants (cereals) such as rice, maize and wheat. However, their most weeds are broad-leaved dicotyledonous plants. So, synthetic auxins are used as selective weed killers; whereby if applied in the fields, they are more absorbed by broad-leaved dicotyledonous plants; which increases their metabolism; thereby increasing their growth rate that becomes unsustainable leading to death. However, the narrow-leaved crop plants (monocots) are not affected and continue to grow normally. Hence, they are freed from competition for light, space, water and mineral salts that would be exerted by weeds.

The synthetic auxins used by farmers and gardeners are **selective, simple, cheap** to produce and have a **very low toxicity** to mammals.

4. **Other commercial applications** include;

- ✓ Auxins are used for production of seedless fruits.
- ✓ Ethene is used to promote fruit dropping in plants like cotton, walnuts and cherries.
- ✓ Cytokinins are used to prevent ageing of ripened fruits and products like lettuces. They are also used to control tissue development in micropropagation.
- ✓ Gibberellins are used to delay ripening and ageing in fruits; to improve fruit size and shape as well as to speed up the malting process in beer brewing.
- ✓ Gibberellins and ethene are used to induce flowering but ethene mainly stimulates early flowering in perennial crops.
- ✓ Auxins are used to prevent sprouting and prolong dormancy. Cytokinins are used to keep flowers fresh.
- ✓ ABA is sprayed on fruits to induce fruit dropping so that there's no need to collect the fruits over a long time span.

SUMMARY OF THE COMMERCIAL APPLICATIONS OF PLANT HORMONES

1. Selective herbicides/weed killers

Auxins are used to destroy broad leaved (dicot) weeds from cereal fields; by making them to elongate and hence can't sustain their growth; leading to death

2. Growth promoters

Auxins are used as rooting powder on stem cuttings.

3. Retard growth; which enables plants to be limited to heights easy to harvest and resistant to wind breakage using anti-gibberellins like cycocel leading to dwarfism.

4. Induce flowering

Gibberellins and ethene are used to induce flowering but ethene mainly stimulates early flowering in perennial crops.

5. Induce fruiting
Auxins(IAA) is applied to unpollinated flowers to induce fruiting by parthenocarp e.g. in seedless oranges and tomatoes.
6. Fruit ripening
Ethene is used as a ripener to induce ripening of fruits by applying ethephon like in banana stripped when green.
7. Storage of potatoes and other crops
Auxins are used to prevent sprouting and prolong dormancy. Cytokinins are used to keep flowers fresh.
8. Gibberellins(GA_3) are used in brewing industry for malting by stimulating the production of alpha-amylase in barley which also improves on the quality of the malt.
9. ABA is sprayed on fruits to induce fruit dropping so that there's no need to collect the fruits over a long time span.
10. Retard abscission.
Auxins and cytokinins are used to delay abscission by counteracting the effects of abscisic acid and ethene.

INTERACTION (CORRELATION) OF PLANT GROWTH SUBSTANCES

In nature, plants have a cocktail of growth substances and the balance of their concentrations leads to the balanced effects as discussed herein;

- (a) **Synergism**; is the control where the combined effect of two or more hormones/substances is much greater than the sum of their separate effects. In synergism, each hormone complement each other's effect to give a greater response. Therefore, it is the hormonal interaction to promote each other's activity/effect. Such hormones/substances are referred to as being **synergistic**.
E.g.
- ✓ In cell division and differentiation, cytokinins promote cell division only in the presence of auxins, gibberellins and auxins play a role in the cambium development.
 - ✓ In shoot growth, gibberellins promote elongation of stem, leaves and petioles in presence of auxins.
- (b) **Antagonism**; is the control where (two) substances/hormones have opposite effects on the same process. In other words, is a type of hormonal interaction where hormones inhibit each other's activity. Such substances are **antagonistic**.
E.g.
- ✓ ABA stimulates bud dormancy while GA_3 breaks it.
 - ✓ Auxins inhibit abscission while ABA promotes it.
 - ✓ Auxins promote shoot and root growth while ABA inhibits it.

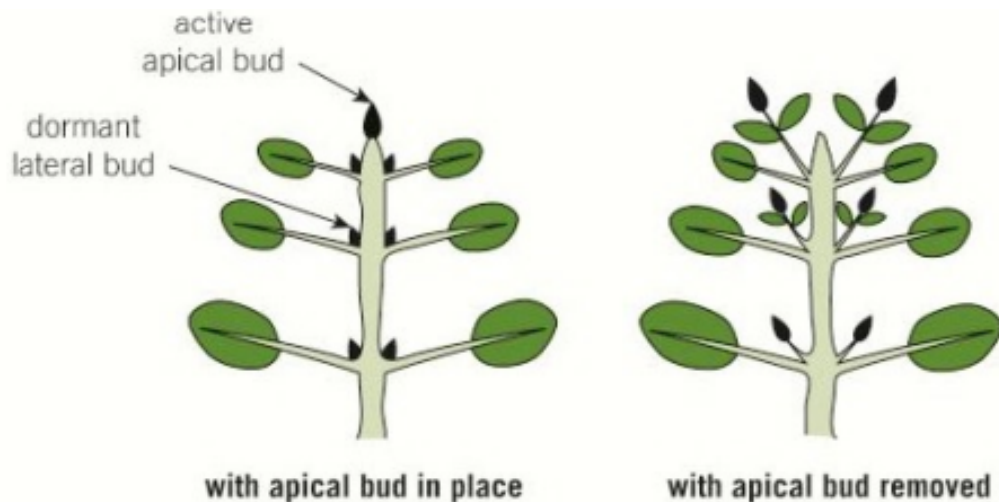
SOME OF THE AFFECTED PLANT PROCESSES

1. APICAL DOMINANCE

This is the phenomenon where the presence of a growing apical bud inhibits the growth of lateral buds. The apical bud is the source of auxins and gibberellins which diffuse to the zone of cell elongation and cell differentiation. These hormones attract cytokinins and mineral salts from the roots to the apex and food like sucrose exported from the leaves to the apex; which promotes growth of the apical bud, leaving the lateral buds dormant due to low auxin and gibberellin levels but with high ABA inhibitor.

Removal of apical bud by pruning promotes growth of lateral buds. This is because cytokinins and mineral salts from roots and food from leaves are channeled to the lateral buds; which makes them active to produce their own cytokinins, auxins and gibberellins; which in turn inactivate growth inhibitor ABA.

Illustration



HINT: Correlation is a phenomenon where one part of a plant controls another under the influence of growth substances.

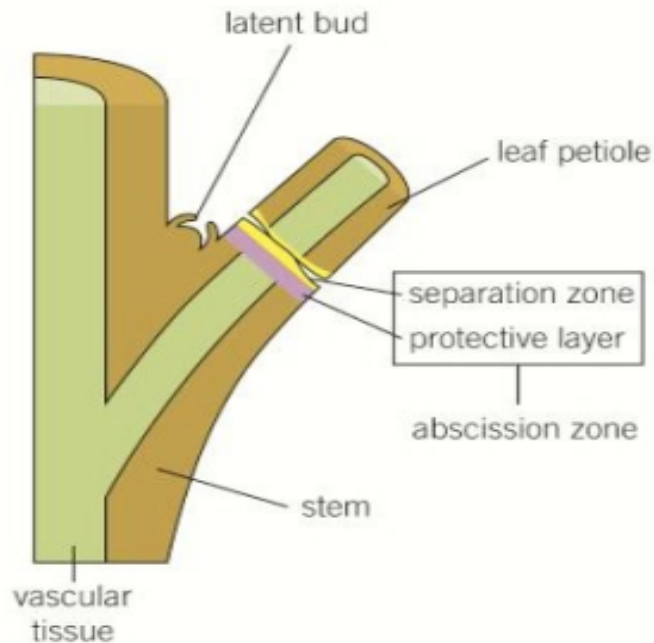
2. ABSCISSION

This is an organized shedding of part of a plant usually a leaf, unfertilized flower or fruit. It occurs at the base of the organ in the abscission zone; where a layer of living cells separates due to breakdown of their middle lamella or cell wall forming abscission layer and a protective layer to prevent infection and desiccation usually at the onset of winter or dry season due to the rate of transpiration in leaves; finally, vascular strands/bundles are broken mechanically by the action of wind. The corky outer layer of the periderm forms a scar; the raised dot indicates a former vascular bundle. This is mainly due to decreased auxin levels and increased ABA levels; therefore, are **antagonistics**.

NB:

- ✓ In fruit ripening both ABA and ethane increase but auxin levels decrease.
- ✓ Abscission of a leaf leads to abscission of petiole due to lack of auxins.

A diagram showing the abscission layer in a leaf stalk



3. POLLEN TUBE GROWTH

This is due to germination of the pollen grains; forming a pollen tube through the style and ovary; which produce more auxins for fruits to set by retaining the ovary after fertilization; which allows fruit development; otherwise abscission occurs.

4. PARTHENOCARPY

This is the development of fruits without fertilization. A few fruits like banana and pineapples proceed to fruit development without fertilization. Usually this is because their ovaries have very high auxin concentration or artificially induced by application of auxins. The same effect is caused by gibberellins with support of cytokinins; which lead to development of the embryo with accumulated food reserves.

5. FRUIT RIPENING

This is brought about by the process of senescence; accompanied by the burst of respiratory activity (climacteric) due to action of ethene in association with ABA that leads to abscission.

6. SENESCENCE

This is the gradual decline in the normal functioning of an organism leading to death.

Characteristics

- Continued degradation of cells.
- Chemical changes in protein molecules.
- Increase in metabolic failure.
- Frequent mutation hence protein synthesis mistakes.
- Decrease in functional capacity.

Types of senescence

(a) Whole plant senescence

During this one, the whole plant (stem and roots) die after seed production like in beans, rice, maize and bamboo.

(b) Progressive/sequential senescence

During this one, older parts of the plant die first, then next ageing ones and later the whole plant like in perennial plants,

(c) Shoot senescence

Here, only the part(s) above the ground die every year; underground parts survive and produce the new plant at the onset of favourable environmental conditions. E.g. In yams, bananas and irish potatoes.

(d) Deciduous senescence

Here, leaves of the plant shed at the same time. E.g. in temperate trees in autumn season.

ADVANTAGES OF SENESCENCE

- (i) It allows old inefficient organs are replaced by new, young and developing organs like leaves, buds, flowers and fruits.
- (ii) It enables more nutrients are withdrawn from old aging plants to the developing buds in need of energy.
- (iii) Bud dormancy and leaf fall are adaptations to unfavourable conditions like winter and drought by decreasing metabolic needs and transpiration rates.
- (iv) It allows Leaf fall which enables recycling of nutrients which maintains growth and development.

LIGHT AND PLANT GROWTH AND DEVELOPMENT

The effect of light on development is known as **photomorphogenesis**.

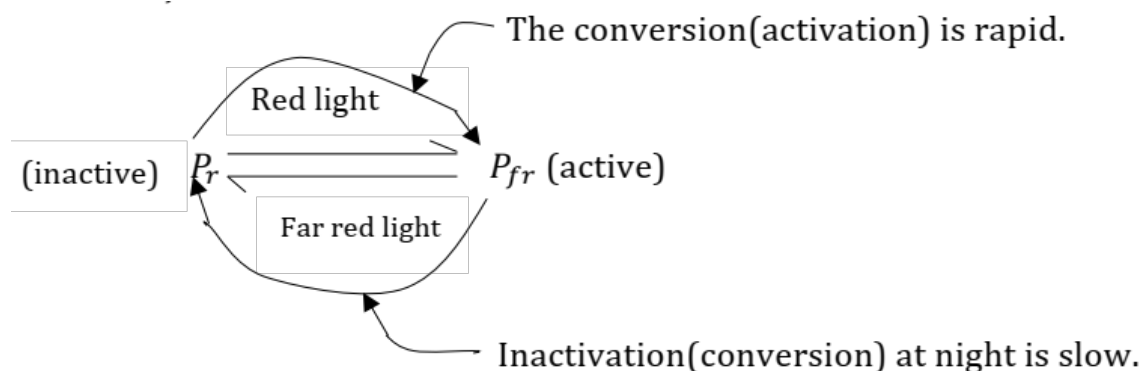
Light affects plant growth and development by influencing essential plant processes such as flowering, etiolation, seed germination, stomatal opening, photosynthesis and plant movements. The light which affects these processes is usually received by photoreceptors concentrated in growing tips called phytochromes. Therefore, these processes are controlled by a phytochrome system in a process involving day length sensitivity known as **photoperiodism**.

THE PHYTOCHROME SYSTEM

It consists of phytochromes. Phytochromes are light-sensitive blue-green pigments existing in two(2) interconvertible forms-that's, phytochrome far red (P_{fr}/P_{730}), which absorbs far-red light within the range of wavelength 700nm to 730nm and phytochrome red (P_r/P_{660}) which absorbs red light within the range of wavelength 580nm to 660nm of visible spectrum. Hence, their interconvertibility depends on the **quality of light**-that's, red light and far-red light.

NOTE: Each phytochrome has a pigment portion attached to a protein.

P_r is a more stable, inactive form that predominates in darkness; during the day, it absorbs red light and rapidly gets converted to P_{fr} . P_{fr} is an unstable, active form which predominates during the day. This is because sunlight has more red light so, its presence **initiates** enzyme activity leading to a biochemical process like flowering and seed germination etc. In darkness, it's slowly converted into P_r , leading to inactivation.



SIGNIFICANCE OF RED LIGHT/PHYTOCHROME FAR-RED

- (i) Promotes germination of seeds like lettuce and fern spores.
- (ii) Promotes leaf expansion in dicots and unrolling in monocots.
- (iii) Promotes chlorophyll development due to conversion of etioplasts into chloroplasts.
- (iv) Inhibits internode growth.
- (v) Stimulates flowering in long day plants.
- (vi) Inhibits flowering in short day plants.
- (vii) Promotes unhooking of plumule.
- (viii) Promotes growth of lateral roots.
- (ix) Promotes greening-that's, synthesis of chlorophyll due to conversion of protochlorophyll to chlorophyll.

PHOTOPERIODISM

This is the response of an organism to changing lengths of the day and night periods. In other words, it is the influence of relative day and night length on activities of an organism. The lengths of the day or night period are known as **photoperiods**.

PHOTOPERIODISM AND FLOWERING

Flowering is mainly affected by photoperiods. As far as photoperiodic requirements for flowering are concerned, flowering plants are categorized into three (3) groups namely;

- (i) Short day plants
- (ii) Long day plants
- (iii) Day neutral plants

SHORT DAY PLANTS(SDPs)

These are plants which only flower when day periods/light periods are/become shorter than the critical length of the day. The nights should also be uninterrupted. E.g. Sugar cane, soya beans, tobacco, rice and strawberry etc. Their flowering can be induced in green houses/growth chambers artificially by using automated blackout systems like blackout curtains/screens or retractable roofs to create artificial short days and lengthen night periods (to provide long, uninterrupted dark periods).

Characteristics of SDPs

- (i) Require short days and long nights to flower.
- (ii) Flower only when days are shorter than critical length of the day.
- (iii) Longer nights are more essential than short days for flowering to occur.
- (iv) No flowering occurs when given short days but long nights interrupted by flashes of light intervals.
- (v) Flower with short days interrupted by intervals of darkness but long nights uninterrupted by flashes of light.
- (vi) Flowering is promoted by increasing phytochrome red or far-red light.
- (vii) Flower during autumn and winter with longer nights than days.

LONG DAY PLANTS (LDPs)

These are plants which only flower when light periods (day periods) exceed (are longer than) critical length of the day. The critical length of the day is about 10 hours. E.g. Wheat, lettuce, clover, cabbage, spinach and barley etc.

Characteristics of LDPs

- (i) Require long days and short nights to flower.
- (ii) Flower only when days are longer than critical length of the day.
- (iii) Longer days are more essential than short nights for flowering to occur.
- (iv) No flowering occurs when given short nights but long days interrupted by darkness intervals.
- (v) Flower with short nights interrupted by light but days remain uninterrupted.
- (vi) Flower in summer with longer days than night periods.
- (vii) Flowering is promoted by high levels of phytochrome far red/red light.

DAY NEUTRAL PLANTS (DNPs)

These are plants which flower after a period of vegetative growth regardless of changing length of day and night periods. That's, they flower in different length of day or night periods hence their flowering is independent of day length. E.g. tomatoes, cotton, garden peas, maize and cucumber etc.

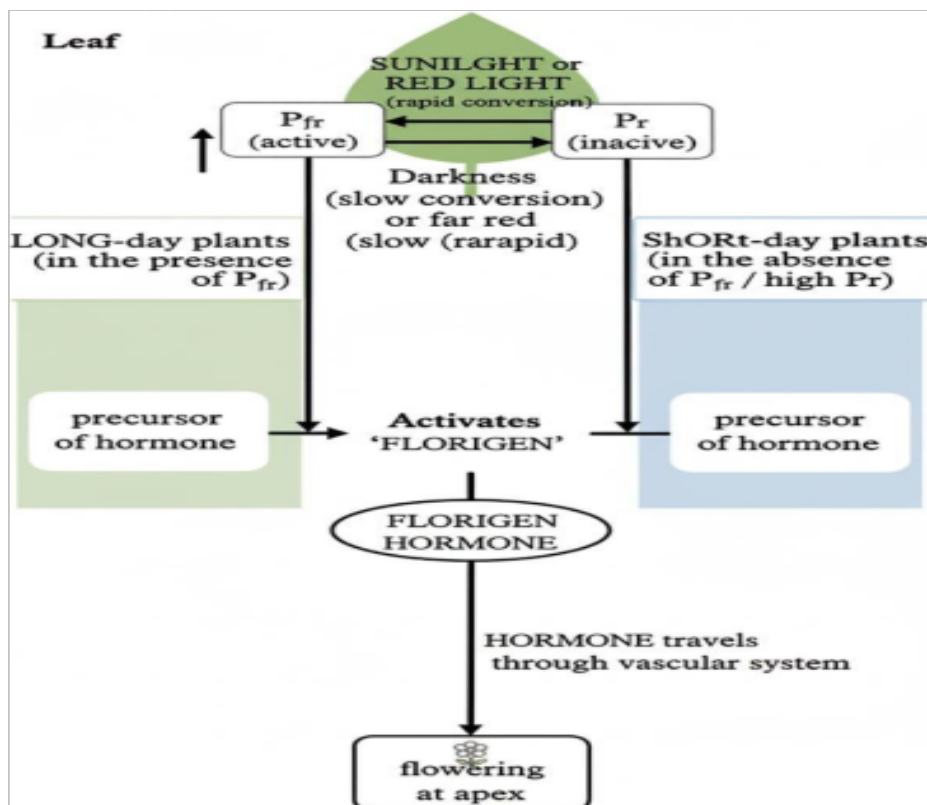
MECHANISM OF FLOWERING

Flowering is controlled by phytochromes as discussed below;

In short day plants, flowering is induced/occurs when the plants are exposed to far-red light at longer periods of nights than the critical night periods. During these uninterrupted longer night periods, the phytochrome far-red molecules in leaves/leaf tissues absorb far-red light and are all slowly converted to phytochrome red; in absence of phytochrome far red in the leaf tissues and high concentrations of phytochrome red stimulate the conversion of the precursor florigen hormone into active florigen hormone; which is actively loaded into phloem tissue of leaves and transported via the sieve tubes to buds by cytoplasmic streaming; where the active florigen hormone stimulates flower buds to develop into flowers.

In long day plants, flowering is induced when the plants are exposed to a prolonged period of red light at longer periods of days than the critical day length/day periods. During these longer day periods, phytochrome red molecules in leaf tissues/leaves absorb red light and are all rapidly converted to phytochrome far-red; (in absence of phytochrome red and) presence of high concentrations of phytochrome far red in leaf tissues stimulate the conversion of the precursor florigen hormone to active florigen hormone; which is actively loaded into phloem tissue of leaves and transported via sieve tubes to buds by cytoplasmic streaming; where the active florigen hormone stimulates the flower buds to develop into flowers.

ILLUSTRATION



NOTE:

- (i) Gibberellins act as red light and promote flowering in long day plants.
- (ii) Gibberellins, cytokinins and auxins can stimulate the production of florigen hormone as phytochromes do.
- (iii) The stimulus is received by leaves and this was proved experimentally by covering the whole plant in a light proof box, leaving only a leaf out and flowers were developed.

APPLICATIONS OF PHOTOPERIODISM IN FLORICULTURE**1. Timing for holidays (off-season) flower production**

Growers manipulate light exposure using artificial lighting (LED bulbs) to extend days or blackout system (opaque curtains/covers) to shorten days in order to induce blooming in photoperiod-sensitive plant species such as Chrysanthemums (SDP) and Easter lilies (LDP) among so as to enable off-season/year-round supply of flowers for holidays, events or exports.

2. Synchronization of crop cycles

Growers manipulate light exposure to plants basing on the knowledge of photoperiodic response to ensure uniform flowering across batches for efficient harvesting and market supply like in cut flowers such as lilies or asters; where uniformity in flowers affects grading, pricing and shelf-life.

3. Extension of market seasons

Allows floriculturists to manipulate light exposure to plants so as to produce high-value, holiday-specific flowers like Easter lilies by varying light treatments depending of the photoperiodic response of a plant species in question. This also speed up the rate of plant growth in greenhouses to meet demand peaks.

4. Bulking up vegetation

Floriculturists expose short day plants to long days by night interruption to delay flowering and prolong vegetative growth phase which enables such plants to bulk up their vegetation before flowering. This in turn enables plants to produce more flowers of high quality with better structural integrity and more appealing to consumers.

5. Controlling storage organs (bulbs and tubers)

The knowledge of photoperiodism is used by floriculturists to promote bulb and tuber formation in floricultural crops by exposing them to short days which enables propagation and timed blooming in greenhouses since many floricultural crops are propagated by means of bulbs and tubers such as lilies, tulips and dahlias etc

6. Preventing bolting in floricultural crops

Floriculturists expose plants to prolonged short days so as to inhibit premature stem elongation and flowering which are usually triggered by long days. This allows plants to flower at the right time which improves the quality of flower flowers.

7. Breeding improvements

The knowledge of photoperiodism allows manipulation of light exposure to plants by the growers to enable development of day neutral cultivars that flower independently of day length, thereby reducing energy costs and enhancing reliability.

ETIOLATION

Plants are sensitive to lack of light in their environment. They grow more rapidly in the dark than when they are illuminated. Gardeners at times use this response to “force” growth in plants by growing them dark sheds (nursery beds). However, etiolated plants are thin and pale. They are pale because the plants are deprived of light thus, little chlorophyll development occurs in leaves. So, soon after germination, the plants should be transplanted to open/lit area for proper growth. This is because the slow upward growth of plants on exposure to light allows the resources to be channeled to essential plant processes like synthesis of leaves, strengthening of stems and overall growth due to a fall in gibberellin levels on exposure of the stems to light.

Ideally, in darkness/semi-darkness, plants become white/pale yellow due to lack of chlorophyll (being chlorotic), shoot internodes become elongated and thick, leaves fail to develop normal chloroplast membrane thus, possess etioplasts. In dicots, the epicotyl and hypocotyl elongate, plumule tip becomes hooked, leaves become small and unexpanded. In monocots, mesocotyl elongate, leaves remain rolled up, chloroplasts lack normal membrane thus, are etioplasts; have less supporting tissues thus, fragile, so they collapse/fall easily unless they reach light.

Etiolation allows maximum growth in length with minimum use of carbon reserves to reach light.

NB: Phytochromes are found in plants, bacteria and fungi.

PHOTOPERIODISM AND SEED GERMINATION

In seeds, light is always perceived by photoreceptors, phytochromes like for lettuce seeds. Red light of 580nm-660nm is most effective wave length for germination while far red light of 700nm-730nm causes inhibition of germination. So, germination doesn't occur as long as P_r is still present.

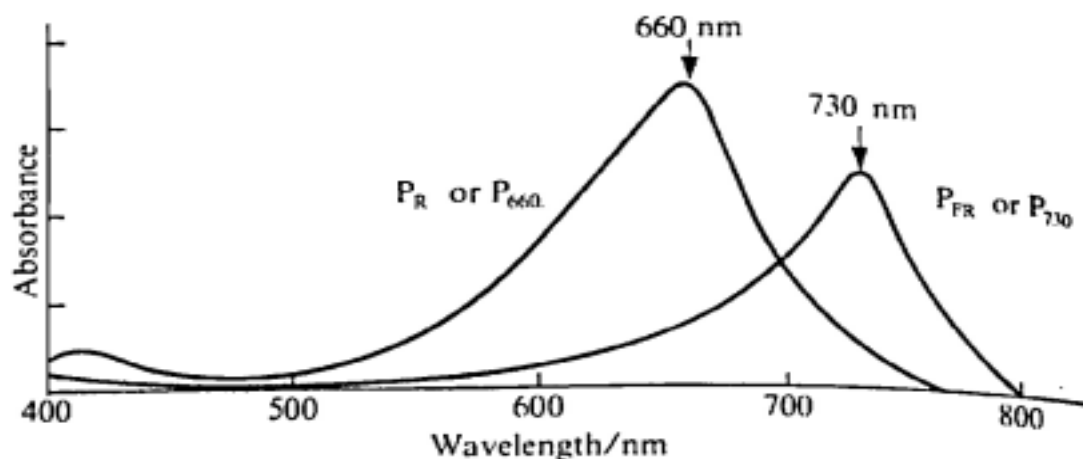
HINT: Far-red light of 730nm is at the end of visible spectrum near infrared.

Only brief exposures of light are enough to affect germination since the effects of red light are reversed by far red light and vice versa. Hence, the last light treatment in an alternating sequence, red or far red is the effective one.

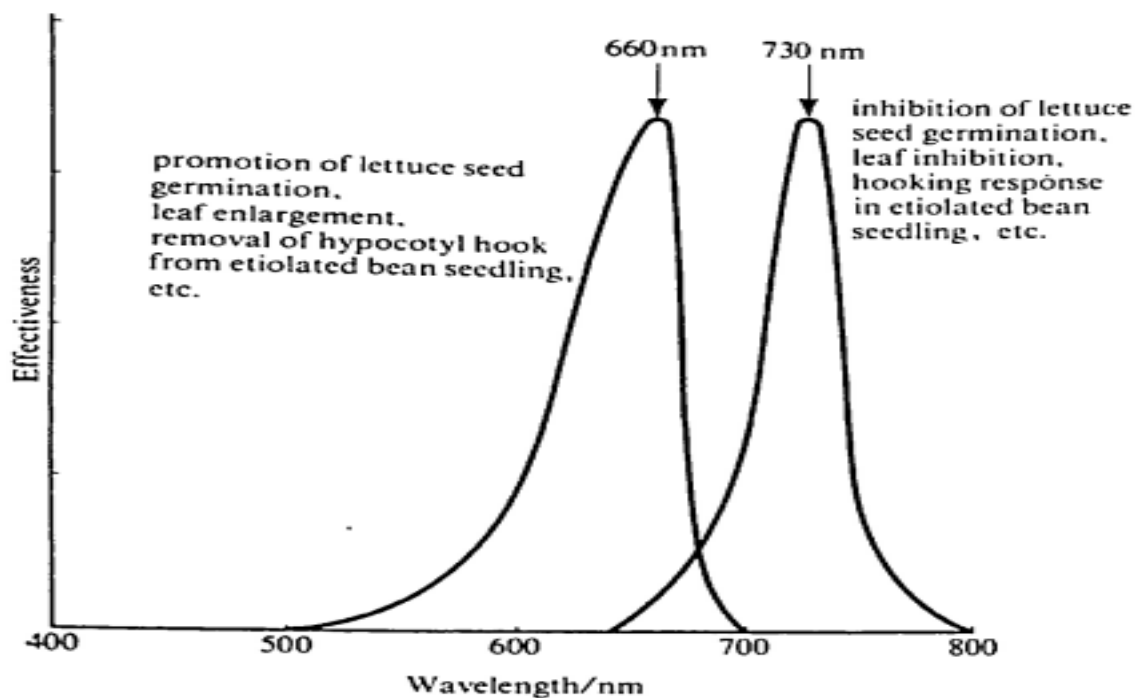
How?

During the day, red light stimulates rapid conversion of P_r to P_{fr} which stimulates germination thus, germination occurs. At night or in the darkness, far red light stimulates slow conversion of P_{fr} to P_r , so no germination occurs. Therefore, exposure to red, far-red, red, far-red etc....; the last one determines what happens. E.g. For seeds exposed to red and then far-red light, the last light quality exposed is far-red light. This implies that the active P_{fr} is converted to inactive P_r thus, no germination occurs.

ABSORPTION SPECTRA OF THE TWO FORMS OF PHYTOCHROME



TYPICAL ACTION SPECTRA OF A PHYTOCHROME-CONTROLLED RESPONSE



The two graphs above are correlated which proves that the effect of phytochromes on seed germination

PHOTOPERIODISM AND DORMANCY

Short days in autumn lead to leaf fall in deciduous plants which breaks dormancy due to exposure of buds to cold that breaks bud dormancy in birch and poplar. They also induce formation of tubers in potatoes.

Long days induce bulb formation.

NB: Oxygen is a vital factor in plant growth since its deficiency leads to retarded root growth.

SEED DORMANCY

A shorter photoperiod causes cessation of growth and the formation of dormant buds. A longer photoperiod breaks the dormancy of seeds. Short days stimulate synthesis of ABA which is translocated to buds leading to temporary suspension of metabolic activity and total inhibition of mitotic activity by inhibiting the synthesis of proteins from mRNA.

Summary of the interaction of plant growth substances on plant

- ✓ Both auxins and gibberellins promote cell division in apical meristems; induce parthenocarpy; stimulate fruit growth; bring about cell enlargement; in return growth. Stimulate flowering in long day plants; cytokinins promote cell division in presence of auxins.
- ✓ Both gibberellins and cytokinins promote lateral bud growth.
- ✓ Auxins promote apical dominance while gibberellins stimulate axillary shoot growth.
- ✓ Auxins promote flowering in short day plants; while gibberellins inhibit flowering in short day plants.
- ✓ Auxins promote femaleness in flowers while gibberellins promote maleness.
- ✓ Auxins promote root initiation from stem or leaf cuttings while gibberellins inhibit development of adventitious roots.
- ✓ Cytokinins promote lateral bud development while auxins inhibit it.
- ✓ Phytochrome red (Pr) inhibits stem elongation while phytochrome far red (Pfr) stimulates stem elongation in long day plants but inhibits it in short day plants.
- ✓ Pr stimulates flowering in short day plants but inhibits in long day plants.
- ✓ Ethylene/ethene inhibits cell elongation while gibberellins promote cell elongation.
- ✓ Absciscic acid stimulates bud dormancy while gibberellins break it.
- ✓ Lateral root growth is inhibited by Pr while it is stimulated by Pfr.

COORDINATION IN ANIMALS

Animals including mammals use two communication systems to coordinate their body activities and these are;

1. **Nervous system**, for nervous coordination via transmission of nerve impulses.
2. **Endocrine system**, for hormonal coordination using chemical messengers called hormones.

Animals usually respond rapidly to stimuli due to nervous coordination associated with electrical communication. The stimuli are either internal or external. Therefore, in this section, we are to concentrate on nervous(neural) coordination in animals since it provides quickest means of in-body communication and effect rapid response to stimuli which increases survival chances.

THE NERVOUS SYSTEM

The nervous system is an extensive system of **neurones** and **synapses** in which electrochemical messages known as **nerve impulses** are transmitted to and from all parts of the body. It collects information about the internal and external environment using **receptors**, processes and integrates it according to past experience and acts upon the information usually by coordinating the organism's activities using **effectors**.

THE NEURONES

These are basic structural and functional units of the nervous system composed of a single nerve cell along which nerve impulses are transmitted-that's, are specialized cells that transmit nerve impulses.

NB: A **nerve** is composed of many nerve fibres enclosed in a connective tissue sheath while a **nerve fibre** is a long axon or dendrite of a neurone.

The 3 terms above-neurone, nerve and nerve fibre are often loosely used interchangeably.

General function of neurones: they transmit nerve impulses rapidly around the body for the body to respond to external or internal stimuli.

STRUCTURE OF A NEURONE

Mammalian neurones possess the following key features;

- (i) **Cell body:** This contains a nucleus surrounded by a cytoplasm. The cytoplasm has numerous endoplasmic reticulum and mitochondria vital for secretion of neurotransmitter substances.
- (ii) **Dendrons:** These are short extensions arising from the cell body. They divide into tiny branches called dendrites. They transmit nerve impulses towards the cell body.
- (iii) **Axon:** This is a single, elongated nerve fibre that transmits nerve impulses away from the cell body. It is cylindrical in shape with a very narrow region

of cytoplasm called **axoplasm** that is surrounded by a cell membrane called **axolemma**.

TYPES OF NEURONES

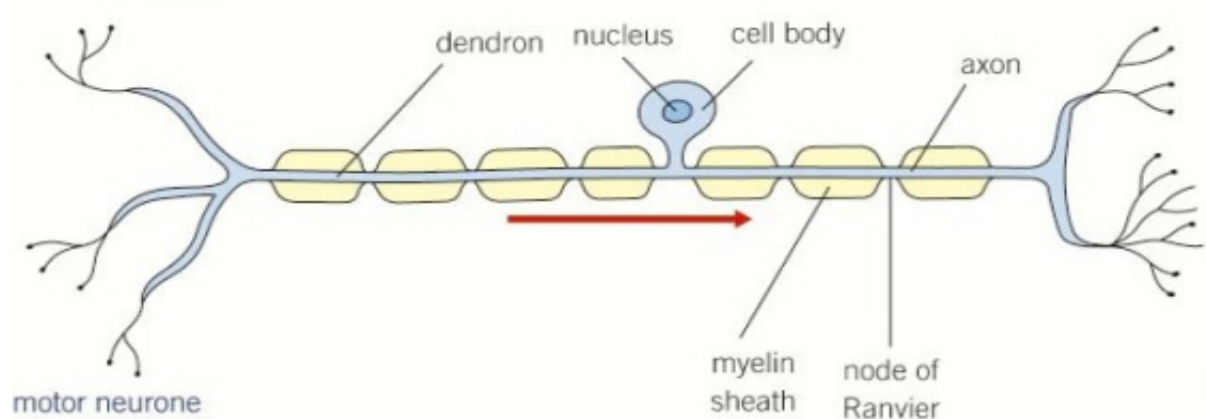
There are three (3) main types of neurones namely;

1. Sensory neurones/afferent neurones
2. Relay/intermediate/internuncial/interneurone/connector neurones
3. Motor neurones/efferent neurones

The sensory neurones

These have one Dendron, cell body aside and one axon. They conduct nerve impulses from receptors to the central nervous system-brain or spinal cord.

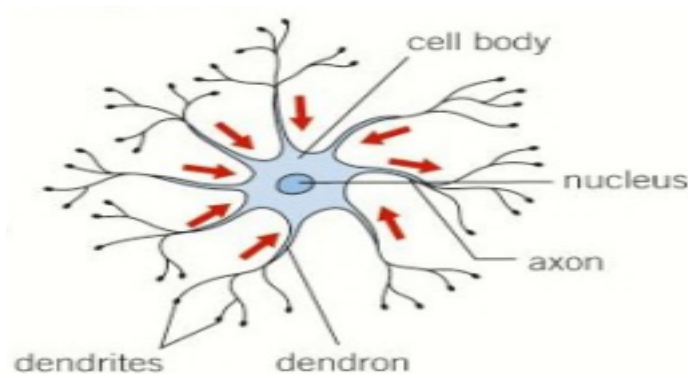
Structure of a sensory neurone



The relay neurones

These have 2 or more short axons and dendrons. They conduct nerve impulses within the central nerve system, linking the sensory neurones to the motor neurones. They enable the brain to process a vast number of information thus, coordination of complicated activities.

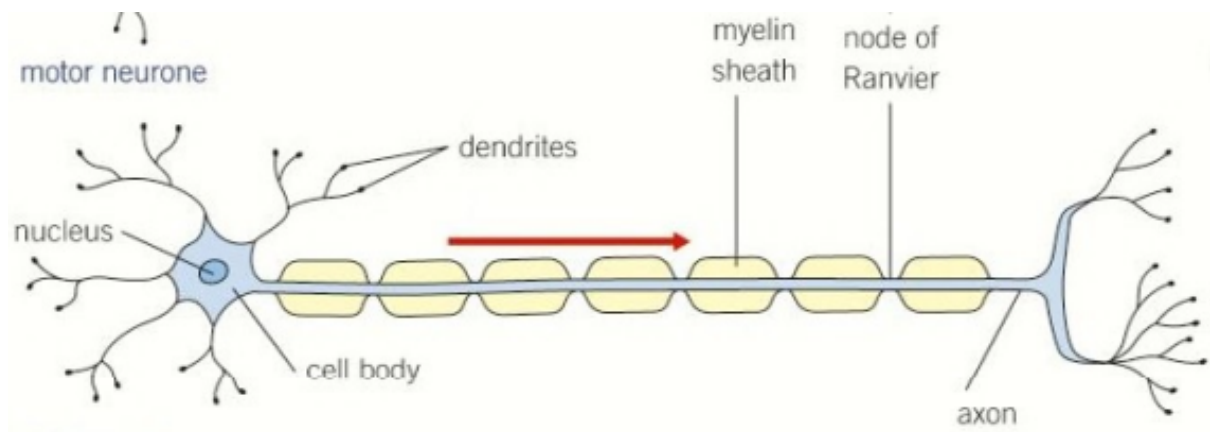
Structure of a relay neurone



The motor neurones

These have one long axon, many short dendrites and one cell body at the beginning. They conduct nerve impulses from the central nervous system to the peripheral effectors.

Structure of a motor neurone



Functions of parts of neurones

- **Dendrites and dendrites**; transmit impulses to the cell body.
- **Axon(axoplasm)**; transmits nerve impulses.
- **Cell body**; contains cellular organelles like nucleus and controls activities of neurone.
- **Schwann cell**; produces myelin sheath.
- **Myelin sheath**; speeds up nerve transmission due to salutatory conduction, insulates and protects the axon from mechanical damages. The uncovered parts of the axon in myelinated neurons are called **nodes of Ranvier** and it's where nerve propagation occurs.
- **Synaptic knobs**; release neurotransmitter substances/chemical transmitters.
- **Axon branch**; supports cell body.

NOTE: Neurones, receptors and muscle cells are excitable cells/units because the value of their resting potential changes. Other cells are non-excitable cells because their resting potential/membrane potential is constant.

ADAPTATIONS OF NEURONES TO THEIR FUNCTIONS

- Have **synaptic knobs** at the terminal end that release neurotransmitter substances for synaptic transmission of nerve impulses hence, communication with other cells and effectors.
- Some have **myelin sheath** for faster transmission of nerve impulses by salutatory conduction, insulation and protection against mechanical injuries.
- Have accessory cells like **Schwann cells** for production of myelin sheath.
- Some have **neuroglia cells** like in brain that protect and support nerve cells and for healing them after damage.
- Have a **cell body** with a nucleus that controls their activities.

- (vi) Have **long axons** for transmission of nerve impulses away from the cell body over long distances.
- (vii) Have numerous **Nissl's granules** for synthesis of proteins like enzymes required for impulse transmission.
- (viii) Have **numerous mitochondria** especially in the terminals that provide much energy in form of ATP required for the release and synthesis of neurotransmitters at the synapse for efficient synaptic transmission as well as pumping of ions across axon membrane to maintain resting potential.

PROPERTIES OF NEURONES

1. Excitability/stimulation

NERVOUS TRANSMISSION

Neurones transmit nerve impulses by temporarily changing the voltage/potential difference across their axon membranes. Thus, nervous transmission is due to **ionic exchange** but not electron flow as it is in electricity.

THE NATURE OF A NERVE IMPULSE

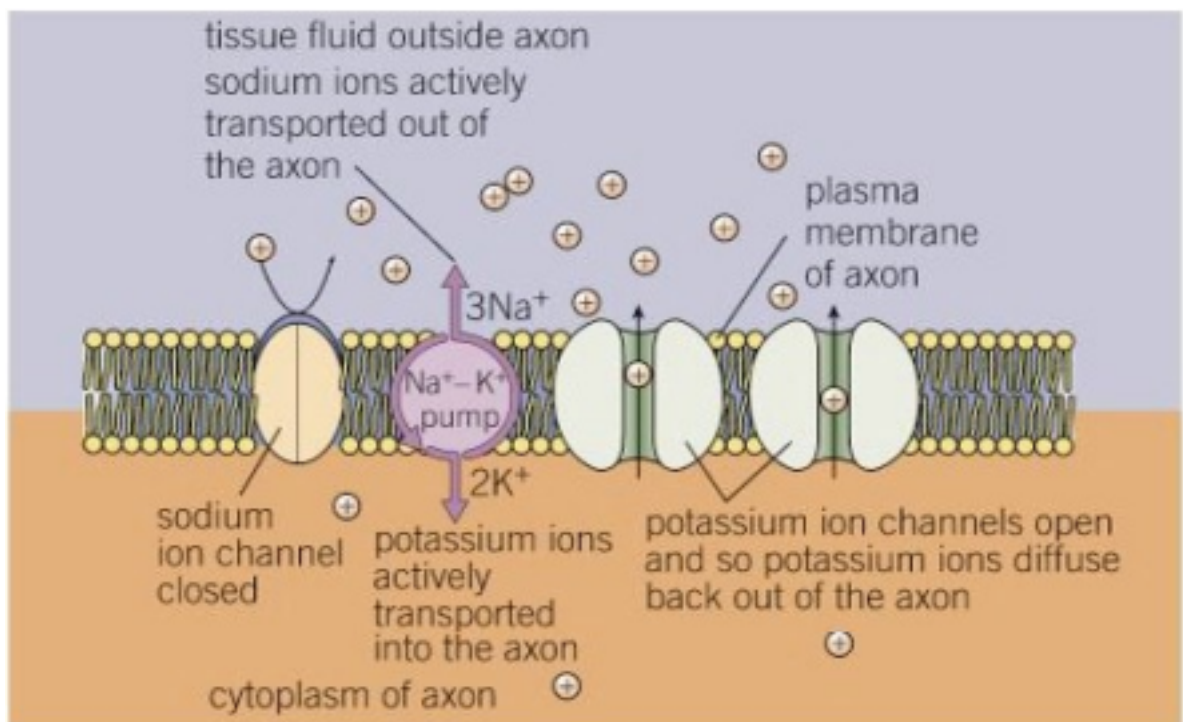
A **nerve impulse** is an action potential propagated/passed along an axon as a wave of depolarization with one portion of axon getting repolarized as the next portion becomes depolarized. Therefore, the axon membrane switches between two states of **resting potential** and **action potential** during nervous transmission.

(a) Resting potential

This is an electric potential difference which exists (is maintained) between the inside and outside of an axon membrane when not transmitting a nerve impulse (at rest). At resting state, it is approximately -70mV .

This is maintained by the action of sodium-potassium pump (mechanism) which uses energy from ATP to pump out 3 sodium ions with intake of 2 potassium ions, opposing the passive diffusion of ions along their electrochemical gradient; with the axon membrane being more permeable to potassium ions than sodium ions. So, potassium ions diffuse back out of the cell freely while sodium ions diffuse back very slowly since sodium ion gates are closed; leading to a net loss of potassium ions contributing to a net negative charge in axon (inside of the axon membrane).

Illustration



NOTE:

- (i) Without active transport, the ionic movement would reach equilibrium leading to no potential difference across the axon membrane.
- (ii) The channel proteins in axon membrane allow diffusion of ions, have gates which can be open or closed. They are also called voltage-gated channels because are sensitive to potential difference/volts.
- (iii) Only some potassium ion channels have gates; so, some potassium ion channels are always open while sodium ion channels open only during action potential. Therefore, resting potential is largely determined by potassium ions electrochemical gradient since chloride ions and organic anions are ignored, and the axon membrane impermeable to calcium ions inside the axon.
- (iv) At rest/ in resting state, the axon membrane is described as being **polarized** because the membrane maintains different electrical charges on the two sides.

How?

At first, the axon loses more sodium ions than potassium ions gain, then more potassium ions loss by diffusion than sodium ions gain which leads to a negative charge inside compared to the outside.

(b) Action potential

This is the potential difference that occurs across the axon membrane on adequate stimulation/arrival of a nerve impulse; where the inside becomes positive relative to the outside.

In other words, it is the polarity change that occurs from negative inside value ($-70mV$) to the positive inside value ($+40mV$) of an axon when adequately stimulated. This is initiated by the arrival of nerve impulses or stimulus application at a point on an axon, causing sudden and brief opening of sodium ion gates leading to **slight depolarization**; which causes sodium ion gated channels sensitive to potential difference changes across the axon membrane to open; increasing the permeability of the axon membrane to sodium ions by positive feedback; leading to rapid diffusion of more sodium ions in axon (sodium ion influx), down their electrochemical gradient; cancelling out the resting potential; making the inside more positive relative to outside, the membrane becomes now depolarized; Thus, potential difference change from negative to positive. If the threshold value is reached, an action potential is generated which is propagated along the axon.

So, action potentials are caused by very rapid movement of sodium and potassium ions across the axon membrane between the extracellular fluid and axoplasm.

NOTE:

- (i) Depolarization/ action potential generation largely depends on movement of sodium ions.
- (ii) Soon after depolarization, repolarization occurs.

Repolarization is the restoration of the resting potential after stimulation or passage of a nerve impulse. At the peak of action potential, almost instantly after depolarization,

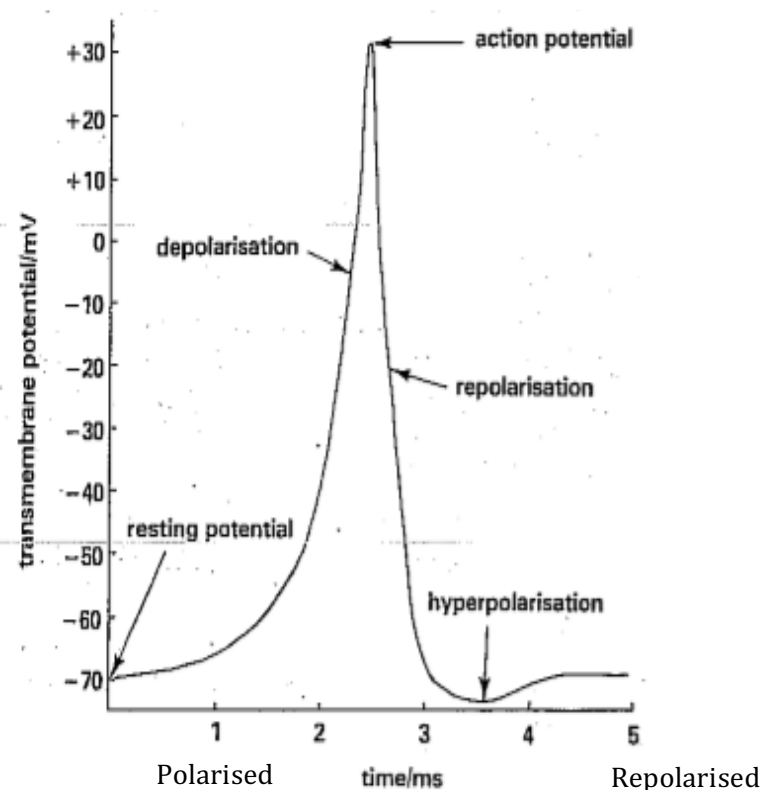
sodium ion gates close, at the same time potassium ion channels open and potassium ions rapidly diffuse down the electrochemical gradient to outside of the axon; making the inside to start becoming less positive again till finally the membrane comes back to the original value, where the inside of the membrane becomes more negative relative to the outside since there are more sodium ions in and more potassium ions outside.

Hyperpolarization; is the when the axon membrane becomes more negative to its resting potential. It is due to delay in closure of all potassium ion gates as compared with sodium ion gates. So, more potassium ions diffuse out of axon leading to slight overshoot (more negative) than the resting potential. Thus, hyperpolarization.

NB: It's the role of sodium-potassium pump and facilitated diffusion of more potassium ions in and more sodium ions out that restores the original resting potential.

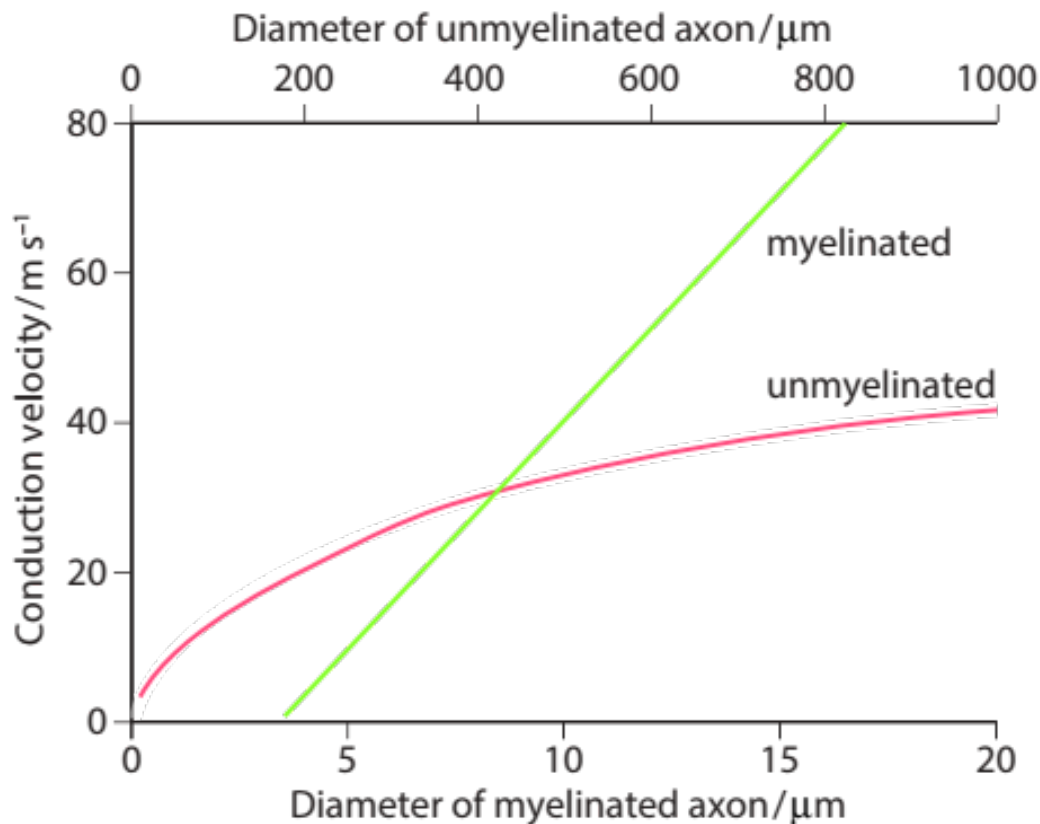
Polarization is the ability to maintain different electrical charges on both sides of a membrane.

Summarizing graph



PROPAGATION OF A NERVE IMPULSE

A graph showing the variation in transmission speed between myelinated and unmyelinated axons of different diameters



NOTE:

- (i) Many annelids (earthworms), molluscs (squids, octopuses), arthropods, prawns and crayfish have giant axons to conduct information ideal for survival fast like for quick escape.
- (ii) Vertebrates have fine axons but faster speed of conduction is achieved by myelination.

TEMPERATURE

At low temperature, the speed of transmission is low; increase in temperature up to 40°C , increases the speed of transmission of nerve impulse and further increase in temperature beyond 40°C , decreases the speed of nerve impulse transmission. This is because at low temperature, low kinetic energy is gained by potassium and sodium ions; so, they diffuse slowly leading to slow rate of ionic exchange thus slow rate of conduction. Increase in temperature up to 40°C , increases the kinetic energy gained by sodium and potassium ions thus, diffuse faster; increasing the rate of ionic exchange (depolarization) during impulse transmission, leading to higher speed of transmission. However, beyond 40°C , sodium-potassium pumps/proteins in the axon membranes are denature which decreases the speed of transmission.

Likewise, increase in temperature increases enzyme activity which increases the rate of breakdown of respiratory substrates/sugars to give respiratory energy thus, more ATP

is provided for active transport of potassium and sodium ions by sodium-potassium pumps, leading to increased rate of ionic exchange during impulse transmission and thus, faster rate of conduction of impulses and the reverse is true for decrease in temperature.

APPLICATIONS

SIGNIFICANCE OF WARM UPS BEFORE PHYSICAL ACTIVITY

Warm ups involve light exercises that gradually increase body temperature within the physiological range; which speeds up nervous transmission by increasing kinetic energy for ionic exchange and thus, faster depolarization and repolarization. Similarly, it leads to increase in enzyme activity which increases the respiratory breakdown of respiratory substrates to provide ATP for the working of sodium-potassium pumps. This improves neuromuscular coordination and quickens reaction time due to efficient impulse transmission from muscles to the brain and back; that increases the nervous system's readiness for intense activity; as a result of gradual activation of sensory receptors in muscles and joints; thereby reducing risks of injuries and preventing strains from sluggish or uncoordinated responses.

THE PRINCIPLE BEHIND THE USING COLD PARKS IN FIRST AID

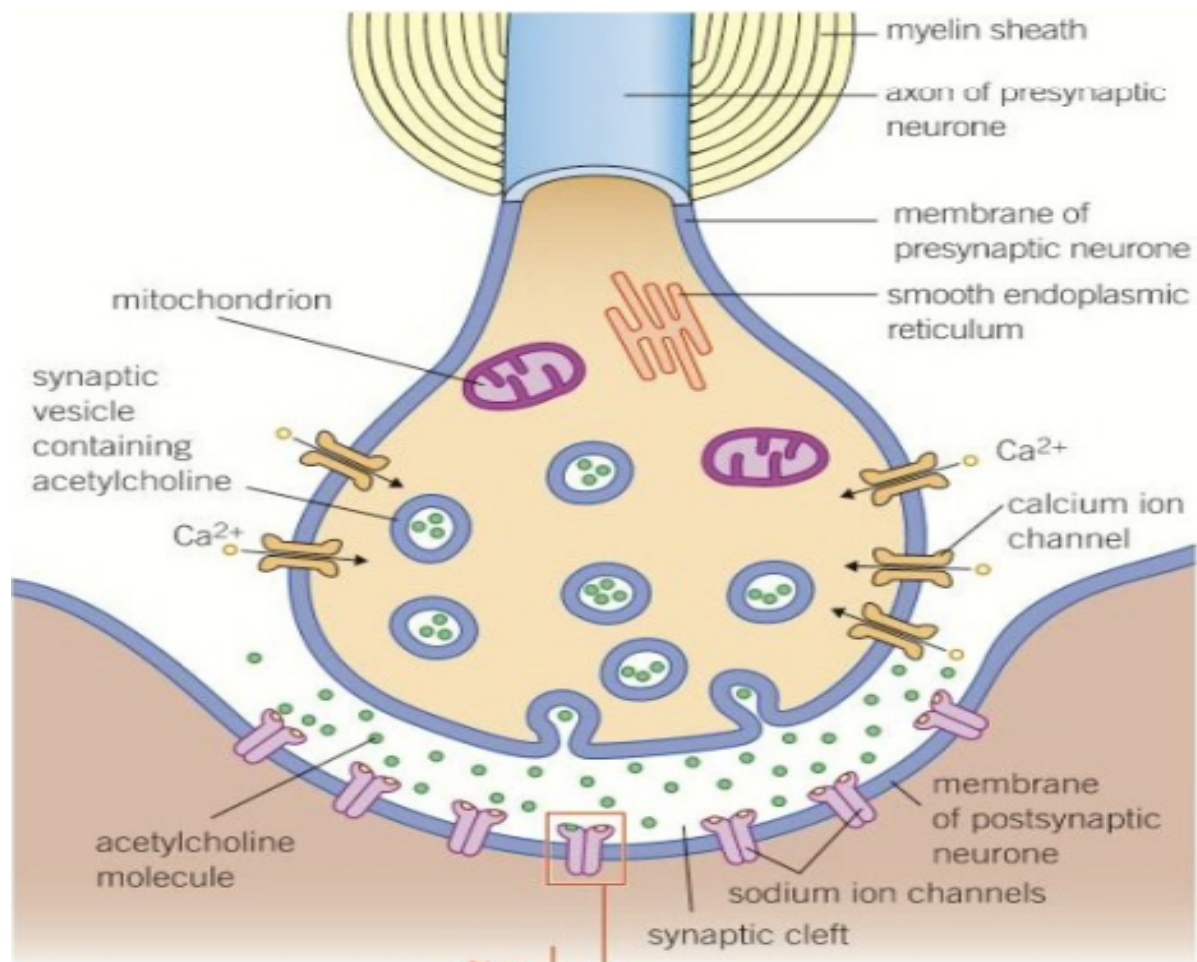
THE SYNAPSES

A **synapse** is a functional link between one neurone and another or muscle/effector with no physical contact between them. So, a synapse can be a neurone to neurone synapse or neurone to muscle synapse called **neuromuscular junction** with a microscopic gap of about 20 nm wide called **synaptic cleft**.

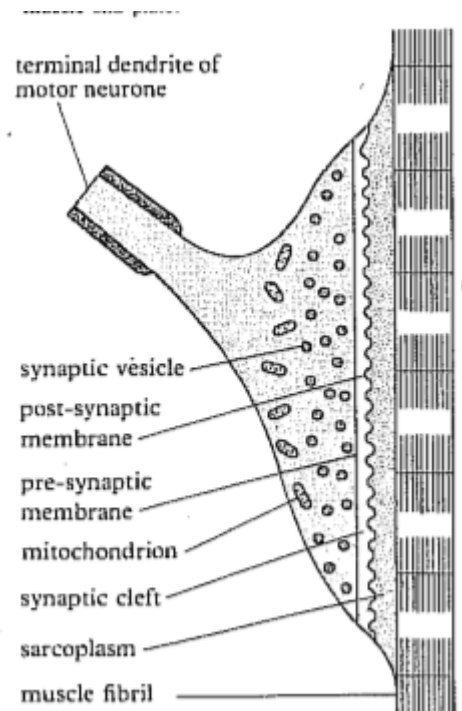
STRUCTURE OF A SYNAPSE

Each synapse consists of a presynaptic knob and synaptic cleft; the neurone to neurone synapse has a post synaptic knob while a neuromuscular junction has postsynaptic muscle fibrils. The presynaptic knob is surrounded by a presynaptic membrane, enclosing a matrix containing numerous mitochondria and synaptic vesicles which contain neurotransmitter substances. The synaptic cleft contains high concentrations of dissolved mineral ions such as calcium, sodium and chloride ions among others and enzymes such as acetylcholinesterase. The post synaptic knob and the post synaptic muscle fibrils are surrounded by postsynaptic membrane which contains specific protein receptor molecules/receptor sites at intervals; the muscle fibrils are within sarcoplasm.

Structure of neurone-neurone synapse



Structure of a neuromuscular junction (neurone-muscle synapse)



NOTE: Neurones that secrete acetylcholine have **cholinergic synapses** while those that secrete noradrenaline possess **adrenergic synapses**.

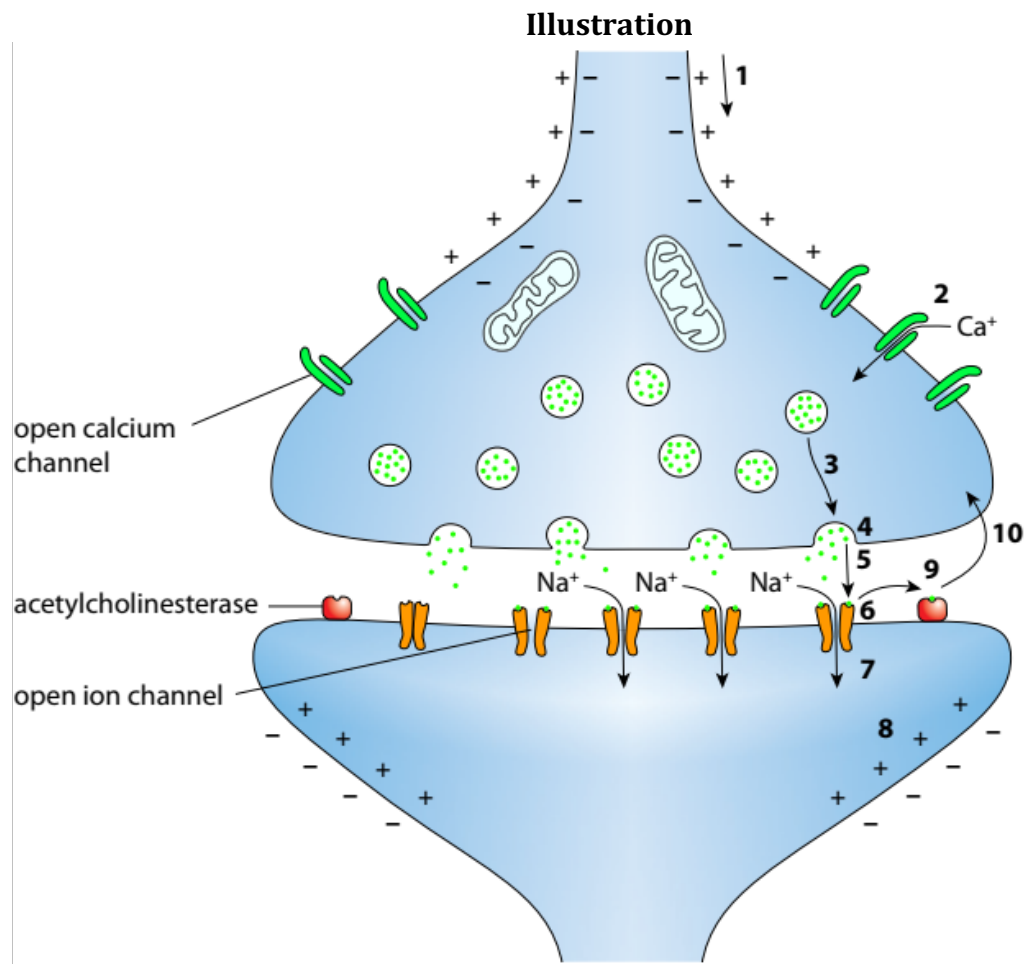
SYNAPTIC TRANSMISSION OF AN IMPULSE

Transmission of a nerve impulse across a synapse is **chemical**.

MECHANISM OF SYNAPTIC TRANSMISSION

Arrival of a nerve impulse at the presynaptic knob increases the permeability of the presynaptic membrane to calcium ions as it gets depolarized and calcium ion channels open; hence, calcium ions diffuse rapidly into the presynaptic knob from the synaptic cleft. (and fuse with the synaptic vesicles) this activates synaptic vesicles to move unidirectionally towards the presynaptic membrane and fuse with it; releasing the neurotransmitter substance like acetylcholine into the synaptic cleft by **exocytosis**. The neurotransmitter substance molecules diffuse across the synaptic cleft and then bind/attach to specific receptor sites on postsynaptic membrane/endplate membrane, causing sodium ion gates/channels to open in excitatory synapse; sodium ions rapidly diffuse inside/into the sarcoplasm; leading to local depolarization, the post synaptic membrane/ muscle endplate becomes depolarized (the outside of the membrane becomes more negative while the inside becomes more positive); the generator potential/endplate potential called **excitatory postsynaptic potential (EPSP)** develops; which builds up to action potential when the threshold value is reached/ exceeded and the action potential/nerve impulse is then fired across the synapse into the postsynaptic neurone/muscle.

The neurotransmitter substance is then hydrolysed immediately from the synaptic cleft by the appropriate enzyme. For example, **acetylcholine** is hydrolysed by acetylcholinesterase/cholinesterase to choline and acetic acid; and the products are reabsorbed actively into the presynaptic knob where acetylcholine is resynthesized from choline and acetyl coenzyme using ATP and cholinesterase enzyme. However, **noradrenaline** is inactivated by monoaminoxidase enzyme that oxidises it.



THE DISTINCTION BETWEEN INHIBITORY AND EXCITATORY SYNAPSES

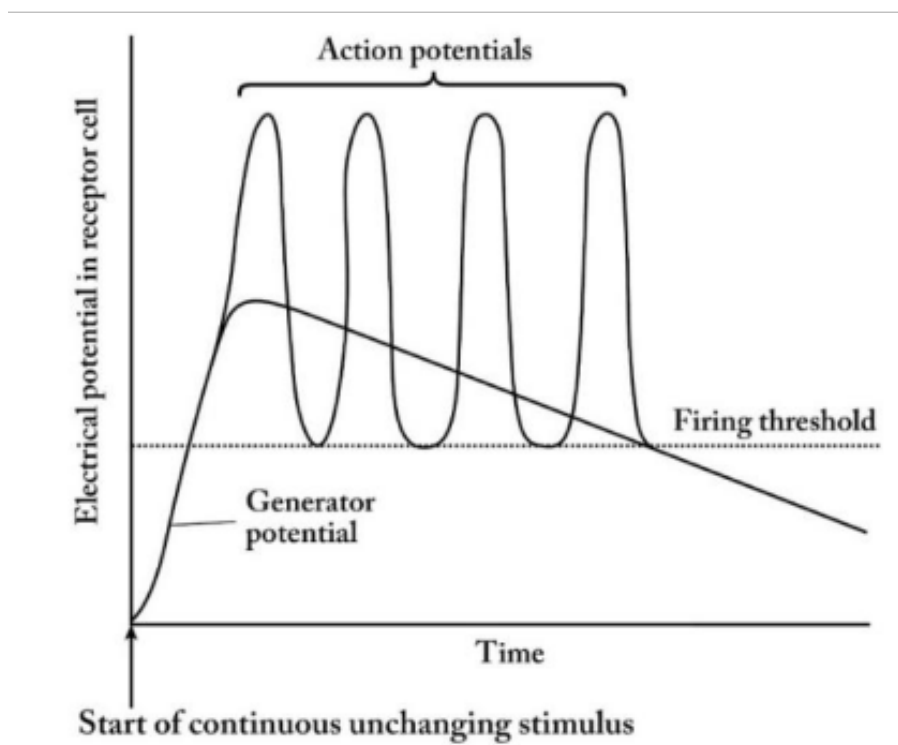
PROPERTIES OF SENSORY RECEPTORS

Receptor cells possess properties which all increase the efficiency of sensory receptors and these include the following;

1. **Various threshold intensities**

Receptors have different threshold **intensities**; where sensory cells with low threshold intensities respond to weak stimuli. As the strength of the stimulus increases, the sensory cell increases the number of impulses fired till a saturation point. At the saturation point (if it can't increase the number of impulses fired,) further increase in stimulus strength will excite sensory cells with higher threshold **intensities**.

ILLUSTRATION



Importance

This extends the range of stimulus for the receptor from low to high level stimulation.

2. **Adaptation**

This is the failure of a receptor to develop action potential due to repeated stimulation. This is because with prolonged stimulation, the membrane surrounding the generator region of the sensory cell becomes progressively less permeable to inward movement/diffusion of sodium ions.

In other words, it is the gradual decrease in the frequency of discharge/firing of impulses due to an intense, prolonged stimulus. It is actually due to increased impermeability of the membrane to sodium ions.

NOTE:

- (i) **Phasic receptors** are rapidly adapting receptors which respond to changes in the stimulus by producing a high frequency of impulses at the moment when the stimulus is switched on or off.
- (ii) **Tonic receptors** are slowly adapting receptors which register a constant stimulus with a slowly reducing frequency. For example, we become insensitive of touch of a clothing on our skin or a man putting on a shirt with sand particles and not feeling them after some time.

Importance

- Provides the animals with precise information about changes in the environment.
- Prevents overloading of the central nervous system with irrelevant and unmanageable information since there's no excessive discharge of neurotransmitters occurs.
- Enables organisms to ignore unchanging background information and concentrate on aspects with most survival value, leading to increased efficiency of nervous system.
- Enables ignoring persistent information.

3. High sensitivity (convergence and spatial summation)

Several sensory cells are connected to and converge on a single sensory neurone to increase; where the cumulative combined effect leads to summation; whereby the generator potentials sum up to reach the threshold value and produce an action potential; yet the effect (generator potential) of a single sensory cell [wouldn't/] can't produce a response (action potential).
E.g. Many rods make synaptic contact with a single bipolar neurone that connects to a single optic neurone leading to retinal convergence like in badgers, owls and foxes which are nocturnal.

4. Spontaneous activity

This is the production of nerve impulses without stimulation. Some receptors produce nerve impulses in sensory neurones in absence of stimulation. Like phantom pregnancy in women where their bodies believe that they are pregnant.

Importance

- It increases sensitivity of the receptor where too small responses (generator potentials) now add to existing potential in the receptor so as to facilitate weak stimuli and generate the action potential when such impulses are produced. In other words, it increases the sensitivity of the receptor by enabling it to make a response to a stimulus that would normally be too small to produce a response.
- It allows the direction of change in stimulus to be registered by this system due to increase or decrease in frequency of impulses (sensitivity/response) in the sensory neurone. For example, the infrared receptors in pits within the face of rattle snake, act as direction finders to locate prey and predators, and they can discriminate increase or decrease of 0.1°C.

5. **Feedback control of receptors**

The threshold of the receptor can be raised or lowered by the efferent (outward) impulses from the central nervous system. This resets the sensitivity of the receptor to respond to different ranges of stimulus intensities. Like the iris of the eye.

Importance: It enables the receptor to extend its range as it can detect wide range of stimuli.

6. **Frequency of discharge**

A receptor generates impulses in afferent nerve depending on the development of a local potential change; whereby the greater the generator potential, the shorter the interval between successive action potentials and thus, the higher frequency of action potential discharge and the lower the generator potential, the longer the interval between successive action potentials thus, less/no action potential evoked (the lower frequency of action potential discharge). Therefore, a weak stimulus generates small generator potential, leading to a low frequency of discharge of nerve impulse in the axon while a strong stimulus generates a large generator potential leading to high frequency of discharge of nerve impulse in the axon.

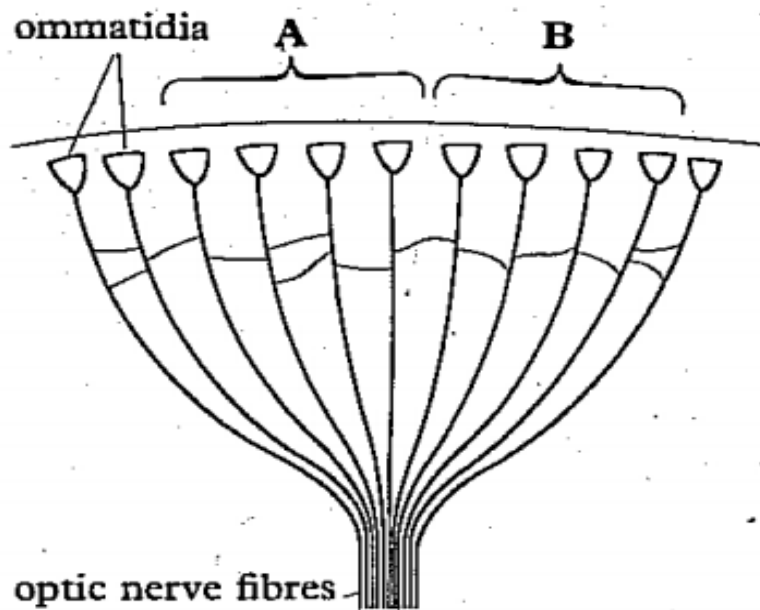
7. **Fusion of stimuli and temporal summation**

When a sensory cell is stimulated repetitively, stimuli are detected separately only if the frequency is low. However, if the stimuli exceed a certain frequency, they fuse into one continuous stimulus. This is because a continuous action potential continual stream is discharged from the receptor; and if the frequency is low, the generator potential “die” down before the next stimulus is received. E.g. When watching a film, one is normally not conscious of the flicker on the screen.

8. **Mutual inhibition/lateral inhibition**

Some sensory cells have synaptic contacts with adjacent sensory units which are inhibitory synapses; such that when several close sensory cells are stimulated simultaneously, they exert an inhibitory effect on each other; so, the frequency of nerve impulse fired/generated by the sensory neurones is lowered, leading to **mutual/lateral inhibition** [making it hard to depolarize the sensory cell membrane]. The nerve impulse frequency is only higher when the sensory/receptor cells are stimulated separately –that’s, when the stimulated sensory cells are apart. This increases acuity or precision. For example, it heightens contrast at light-dark boundaries whereby the further apart the regions of the eye are, the smaller the inhibitory influence over each other. It also sharpens the sense of pitch in the human ear.

Consider the structure of compound eye below;



The compound eye has many packed receptor units called ommatidia. If receptor units labeled A are stimulated with strong beam of light and B with weak beam of light; A and B exert an inhibitory influence on each other by A exerts stronger inhibitory effect on B. so, the light beam on A is not registered as bright as really is but the importance is the difference between the two intensities; they appear greater than really is.

9. Inhibition through efferent neurones

Receptors are inhibited through the efferent nerves by the action potentials from central nervous system where the activities of an animal leads to stimulation of its own receptors and thus, hyperpolarization. For example, the lateral line canals of fish and amphibians have a group of cells that is sensitive to local movement in the surrounding water, which warns the animal of other animals in its vicinity due to the inhibition; whereby when the animal is actively swimming, the lateral line is not excited by its own movements.

10. Precision

The sensory receptors transmit precise information due to their ability to discriminate/resolve stimuli effectively.

Consider;

(a) Precision of the eye (visual acuity)

This is the ability of the eye to resolve two or more stimuli separated spatially as separate entities, leading to high visual acuity. For example, when 2 objects separated in space, when their images fall on the same cone, they can't be passed separately; when they fall on next-door cones, they can't be distinguished. In order to be seen as separate objects, they should fall on different cones with at least one cone in-between. Therefore, the closer the cones, the higher the visual acuity.

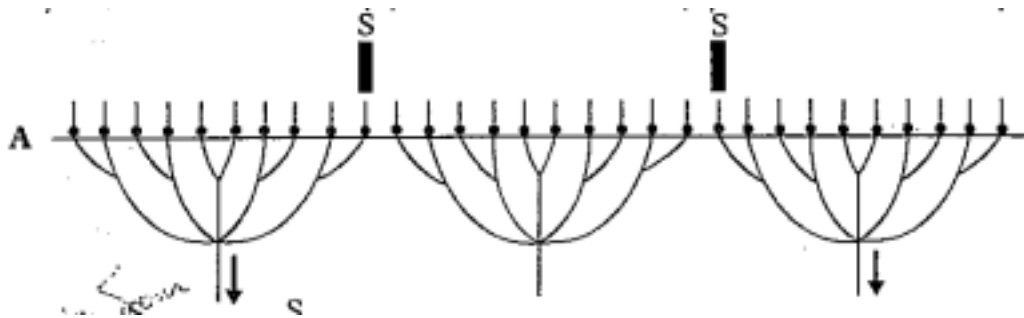
Adaptations of the retina for high visual acuity

- Has most cones concentrated in the fovea (centralis) to increase visual acuity.
- Has a small fovea with thinner and many cones than elsewhere to increase visual acuity.
- In the fovea, each cone has its own bipolar neurone with a single synaptic connection to optic fibre, so no convergence thus, high resolution of stimuli.

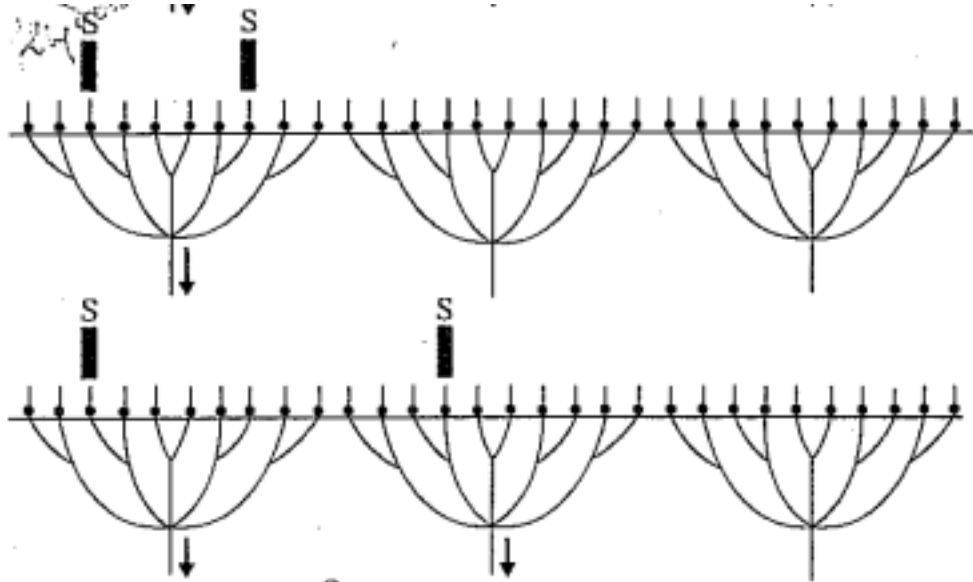
(b) Discrimination of tactile stimuli.

In the skin, there are a large number of tactile receptors which converge onto a single afferent fibre. Upon application of two tactile stimuli like points of a pair of dividers, the two are perceived separately if the distance is great, as the distance reduces, there comes a point when the receptor cells can no longer discriminate the two stimuli and the sensation is that of a single stimulus at a critical distance. That's, a number of tactile receptors sensory nerve for discrimination of stimuli to occur, the stimuli must fall on receptive fields of two distinct nerve fibres separated by at least one in-between.

Illustrations



In the case above, the stimuli can be distinguished

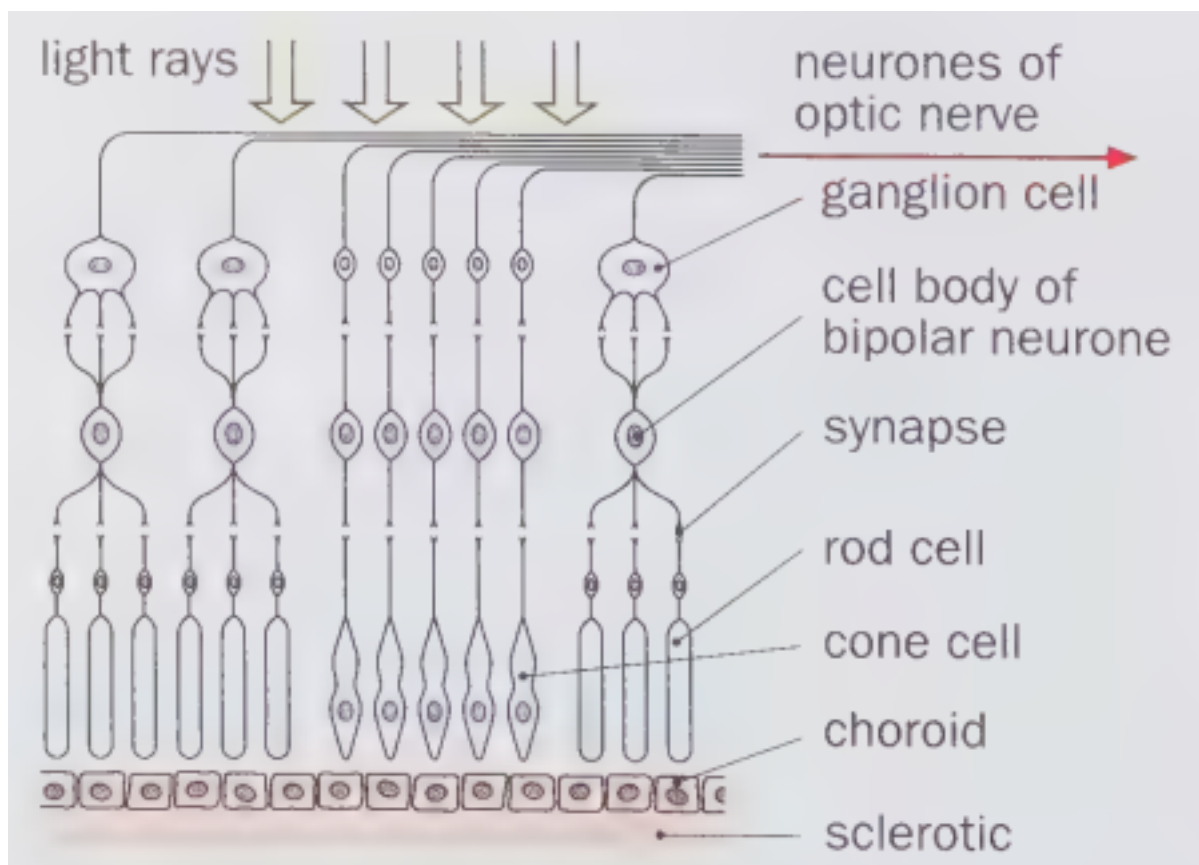


In the 2 cases above, the stimuli can't be distinguished.

STRUCTURE OF THE RETINA

The retina is a very delicate innermost light sensitive layer/photosensitive layer. It contains light/photosensitive cells, rods and cones which are sensitive to dim light and bright light respectively. Rods have rod-shaped outer segments, are distributed all over the retina except at the fovea and responsible for night vision. Cones have cone-shaped outer segments, are concentrated at the fovea only and responsible of colour vision. Rods contain photosensitive pigment, rhodopsin/visual purple and have high visual sensitivity while cones have iodopsin as photosensitive pigment and high visual acuity. The fovea/yellow spot is a slightly depressed light sensitive spot/area on the retina with cones only; blind spot is the area where optic nerve leaves the retina with no rods and cones thus not photosensitive thus the name blind.

Illustrative diagram



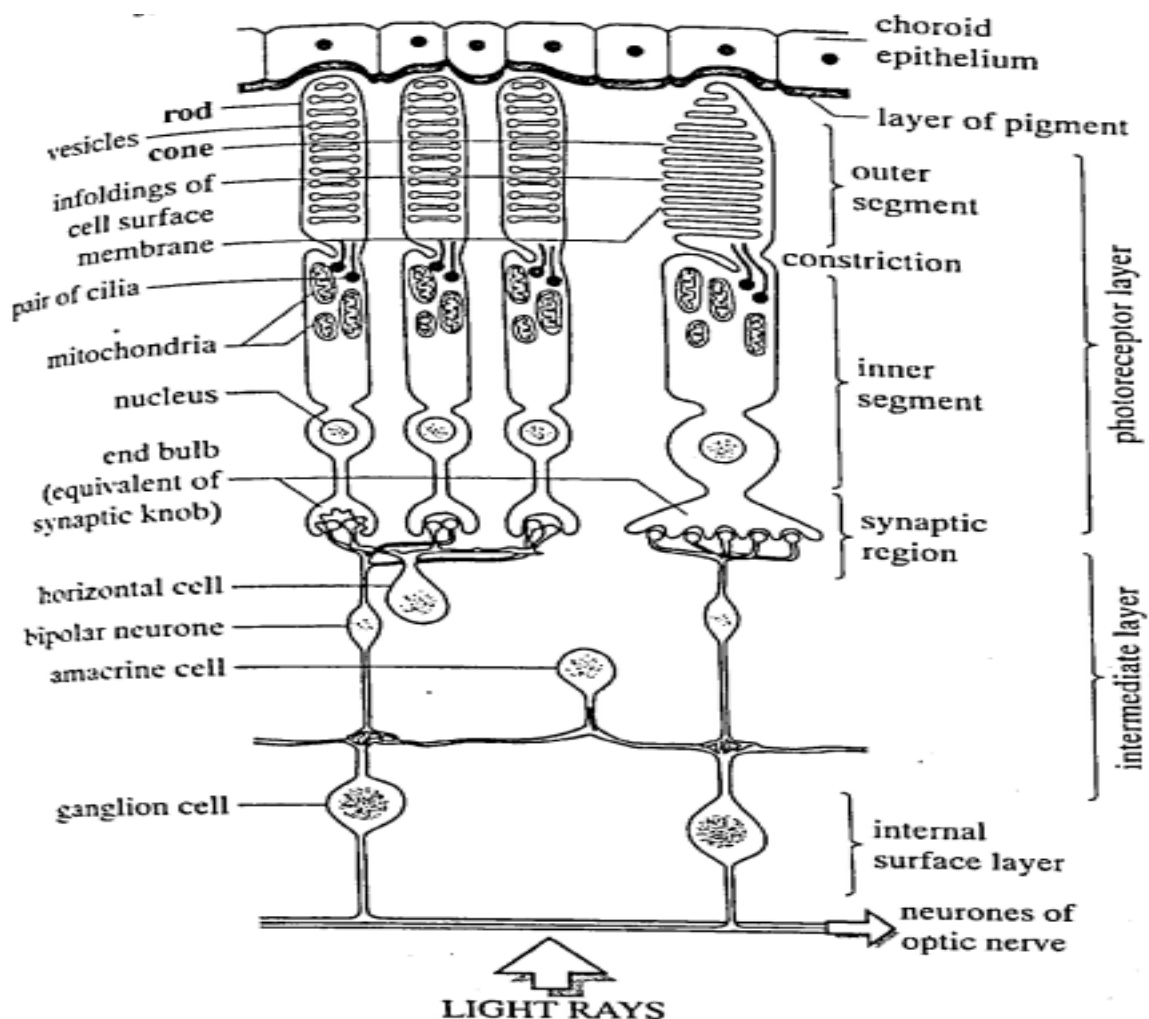
Therefore, from the diagram above, the retina is an **inverted** with photoreceptor cells, rods and cones that are partially embedded in choroid and photosensitive pigments attached to the outer surface of the membrane

STRUCTURE OF THE RODS AND CONES IN THE RETINA

A rod cell consists of four regions-outer segment, inner segment, constriction and synaptic region. The outer segment is rod-shaped and contains flattened vesicles which contain light sensitive(photosensitive) pigment **rhodopsin**. Each rhodopsin is made up of scotopsin and retinene. The constriction separates the outer segment and inner segment, and remains in contact by cytoplasm containing a pair of cilia. The inner segment is a region with cytoplasm containing membraneous organelles like the nucleus, many mitochondria and numerous ribosomes (polyribosomes). At the synaptic region, rod cells are linked to other cells and form synapses with bipolar cells; horizontal cells and amacrine cells link rod cells together.

NB: Cones are different from rods in such a way that *cone cells have cone-shaped outer segments with inner foldings of the surface membrane and each cone has its own synaptic contact with the bipolar neurone*. Similarly, their outer segments contain a photosensitive pigment called **iodopsin** which exists in 3 forms thereby giving rise to 3 types of cones-that's, blue, green and red cones.

A DIAGRAM SHOWING ULTRASTRUCTURE OF RODS AND CONE IN A MAMMALIAN RETINA



NB: Cats have a reflecting layer called *tapetum lucidum* behind the retina which reflects light back into the eye, providing further opportunities for rods to absorb it; leading to night vision and bright reflections (glowing) at night.

THE ROLE OF THE RETINA IN VISUAL PERCEPTION

DISTRIBUTION OF PHOTORECEPTOR CELLS IN THE RETINA

RODS are uniformly/evenly distributed all over the retina but outside the fovea thus, concentrated in the peripheral parts of the retina.

Significance

It enables the outer edge of the retina to be sensitive to low light intensity since it has only rods which are not tightly packed thereby enhancing high stimulation and high sensitivity to dim light.

Why does mammalian eye possess high sensitivity to light?

This is because its retina has rods which show retinal convergence, so when several rods are simultaneously stimulated, their individual generator potentials sum up/summate to develop an action potential in the bipolar neurone thereby increasing collective sensitivity.

HINT: Most nocturnal animals mainly possess rods in their retina for excellent **night vision**.

Effect of the distribution of rods

The distribution of rods is the main reason why we look at the object from an angle but not directly in dim light(night) so that more light falls (be directed/focused) on the rods at the periphery/on the side of the retina in order to form clear images and observe clearly. Also, people usually walk while bent at night in order to focus light rays onto the peripheral retina where rods are located on tilting their heads thereby enhancing vision and avoiding obstacles.

CONES are tightly/closely packed at the fovea/ centralis.

Significance

It makes the fovea to be sensitive to only high light intensity and colour and hence high visual acuity and precision due to high concentration of cones [which synapse with own bipolar neurone].

Why does the mammalian eye possess high visual acuity?

This is because its retina has cones which are densely packed close to each other at the fovea which is the optimum position for receiving enough light; and each cone has its own optic nerve thus sends separate signal/impulse to the brain hence, making it possible to resolve objects close together as separate images are formed.

THE MAMMALIAN EAR

The mammalian ear is a sense organ for hearing and maintaining body equilibrium by detecting changes in the animal's position and head movement. It has mechanoreceptors that are sensitive to changes in sound waves, gravity and displacement.

THE ROLE OF THE MAMMALIAN EAR IN MAINTAINING BODY EQUILIBRIUM

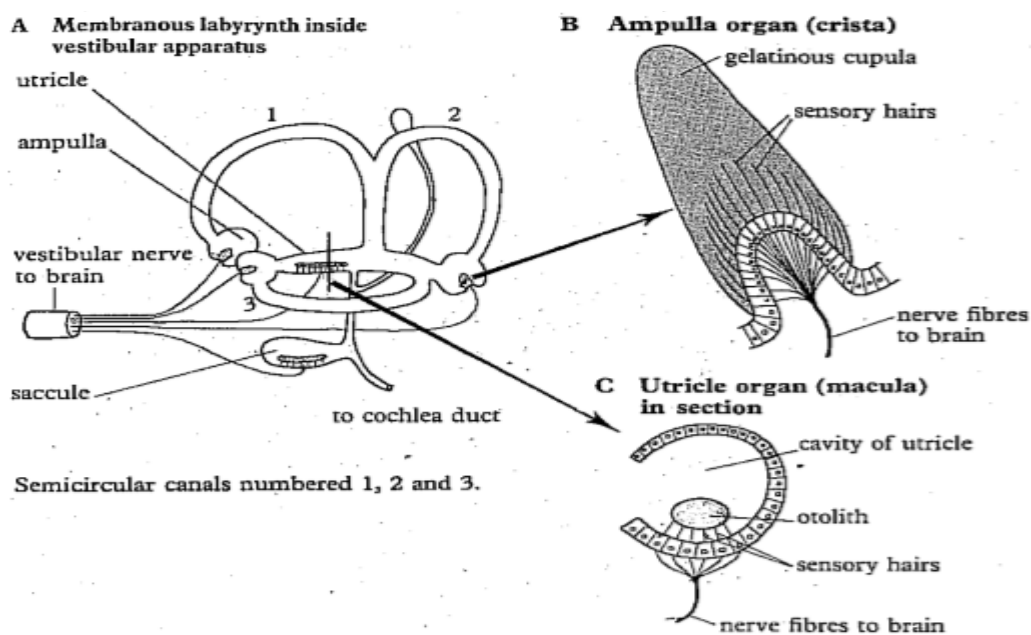
The mammalian inner ear has vestibular apparatus which is responsible for maintaining both **static** (body posture) and **dynamic equilibrium** (balance during movement).

STRUCTURE OF THE VESTIBULAR APPARATUS

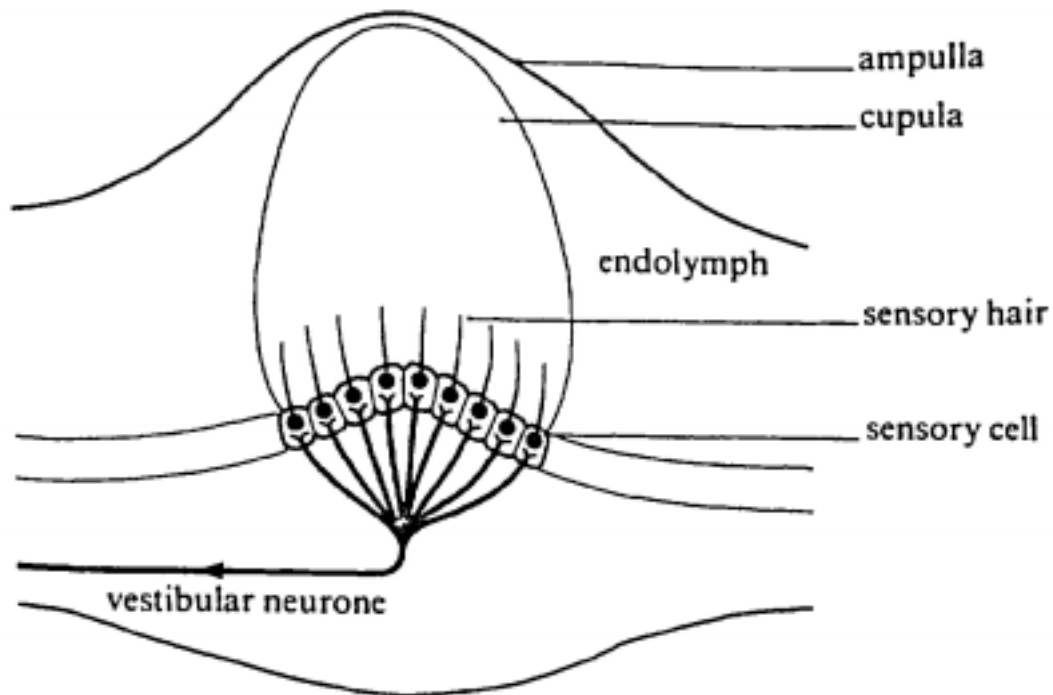
The vestibular apparatus consists of three (3) **semi-circular canals** at right angles, each terminates in a bulge called **ampulla** with **cupula** inside and opens into other cavities known as **utricle** and **saccul**. A **cupula** is a dome-shaped gelatinous structure embedded in endolymph with sensory cells called **cristae**, connected to the vestibular neurone at the base. Each cavity of utricle and saccule contains a gelatinous granule of calcium carbonate called **otolith**, surrounded by an **epithelium** with **sensory cells** (hair cells) called **macula** (plural: maculae) whose free ends are embedded in the **otolith** and connected to the **vestibular neurone**; where all the structures are embedded in an **otoconium** (plural: otoconia).

NB: All the structures of the inner and middle ear are enclosed in the skull bone for protection.

Diagram of the mammalian vestibular apparatus showing the structure of the two main types of balance receptor



An illustrative diagram showing a transverse section of ampulla of a semicircular canal



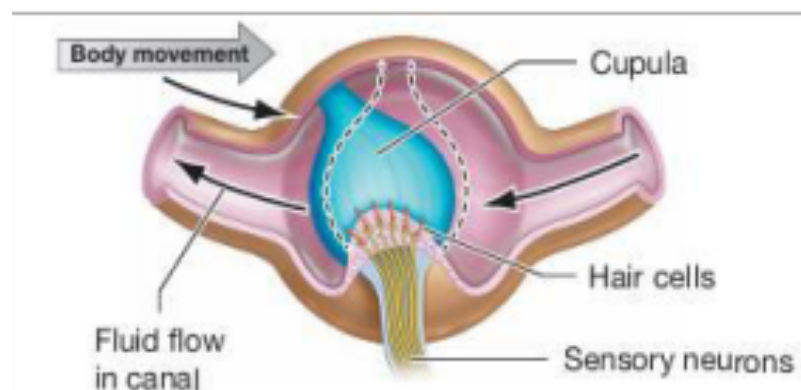
How does the vestibular apparatus maintain body equilibrium?

(a) **Dynamic equilibrium**

The dynamic equilibrium is maintained by the 3 semicircular canals in three planes—that's, vertical canal for upward direction, horizontal canal for backward and forward motion and lateral canal for sidewise movement.

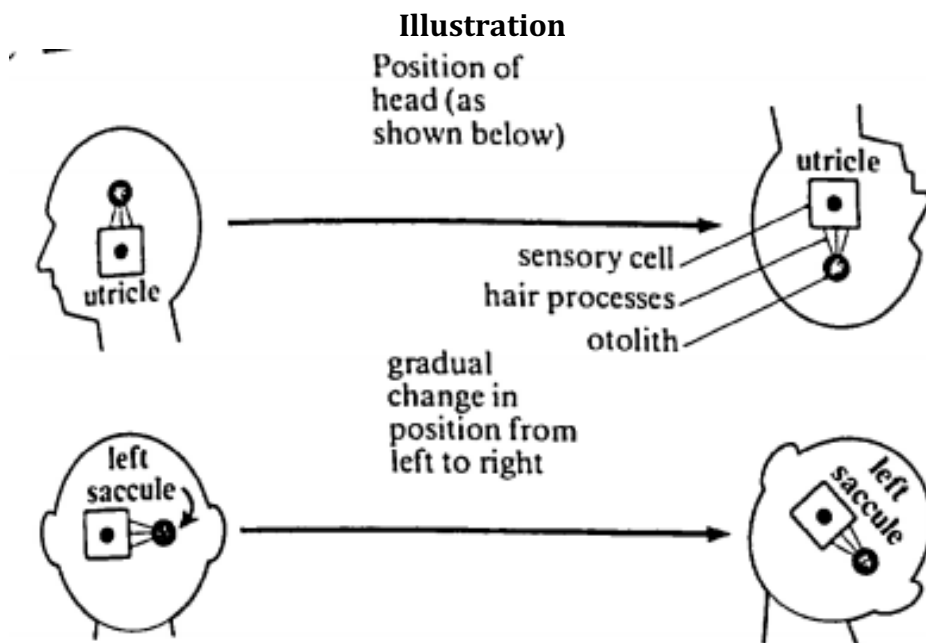
When the head moves in a given direction, the endolymph in the ampulla of one of the semicircular canals moves in opposite direction due to inertia which deflects the **cupula** (in the same direction), stimulating the sensory cells in the cristae on **straining** due to bending of sensory hairs; generator potential develops which builds up to the action potential when the threshold value is attained/exceeded and the nerve impulse is fired along the vestibular neurone to the brain(cerebellum) for interpretation.

Illustration

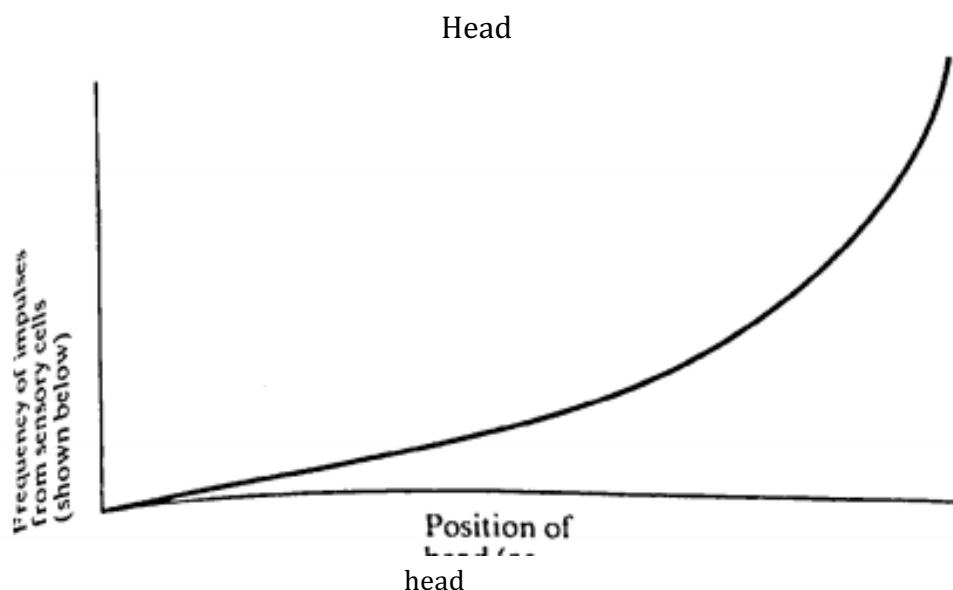


(b) **Static equilibrium** (body posture)

The static equilibrium is maintained by utricles and saccules with receptors in otolith that detect the position of the head with respect to gravity. When the position of the head varies, the pull of gravity over sensory hairs on otolith tilts them accordingly and get distorted, generator potential is developed which builds up to action potential when the threshold value is reached/exceeded and the nerve impulse is fired to the brain via vestibular neurone to interpret the position of the head; such that appropriate instructions are sent to relevant muscles to restore the body balance. When the head is upright, the otolith is positioned on top of the sensory cells and no stimulation occurs since they are not distorted.



A graph showing the effect of the head position on the activity of the receptor cells in the utricle and saccule

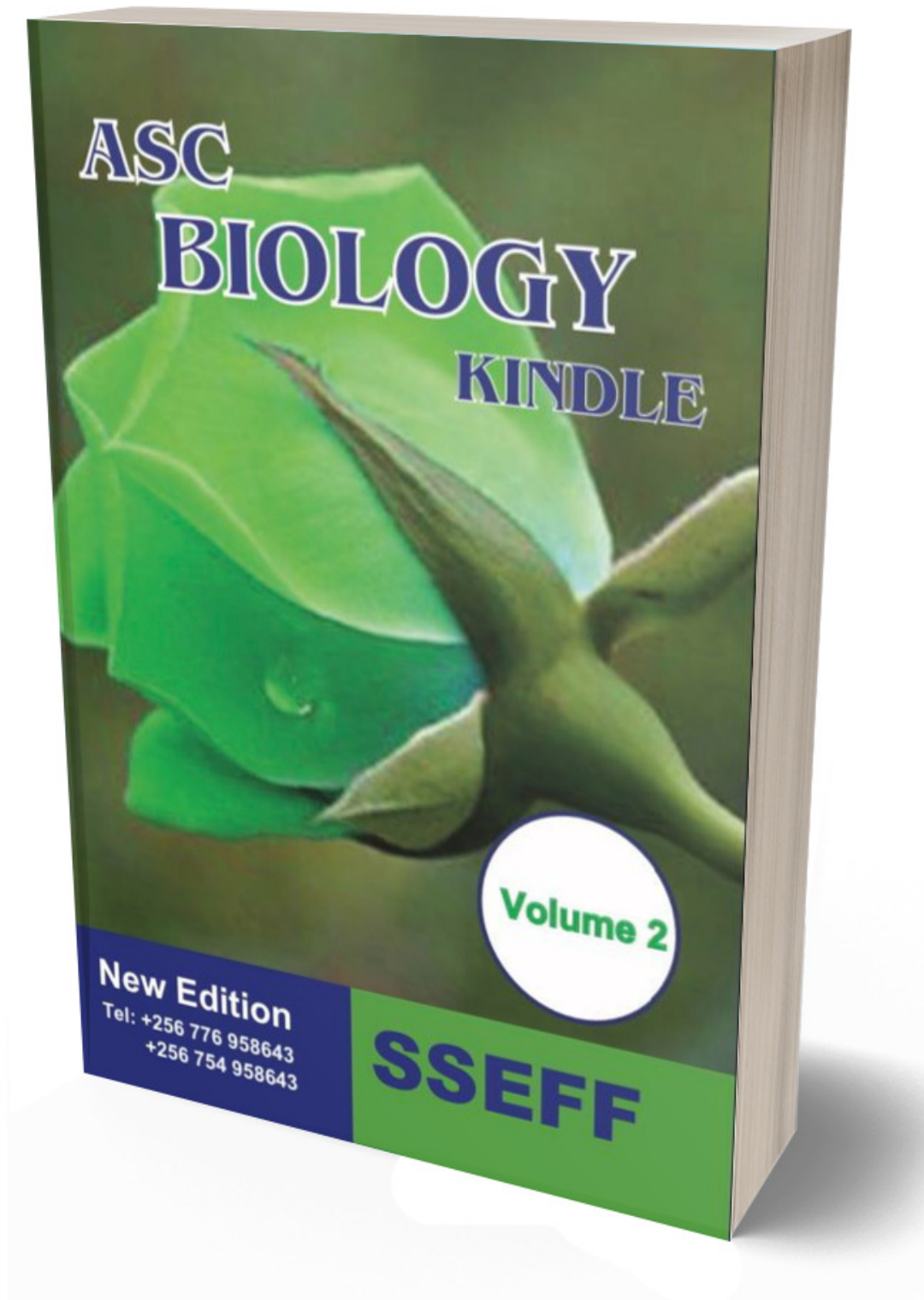


BODY BALANCE SUMMARISED

Balance is brought about by vestibular apparatus consisting of 3 cavities, semicircular canals which open into other cavities **utricle**s and then **saccul**es. Balance is achieved by detecting head movement by semicircular canals, each located in a given plane-that's, horizontal, vertical and lateral plane at right angle to each other, and detection of head position by utricle and saccule. Movement of the head in any plane is detected by sensory cells in the cupula within the ampulla at each end of semicircular canal; for example, on nodding the head semicircular canal in vertical plane is stimulated while shaking the head activates horizontal semicircular canal; when head is moved in any plane the cupula is deflected in direction opposite to that of the head movement due to inertia against endolymph, sensory cells are strained, become depolarized and the generator potential is generated when the threshold value is exceeded, action potential is developed which is fired to brain via auditory neurones which responds by firing motor impulses to appropriate muscles causing them to contract to normalize the head. Position of the head is detected by receptors in utricle and saccule embedded in **calcium carbonate matrix** to form **otolith**; vertical positions are detected by receptors in utricle and horizontal positions are detected by receptors saccule. When there's a change in position of the head, there's a pull of gravity to otolith thus, **differential distortion** of the receptors in otolith, generator potential is developed and when the threshold value is exceeded, action potential is developed and fired to brain via auditory neurones; brain responds by firing motor impulses to appropriate muscles causing them to contract to regain the normal position of the head

NOTE: Proprioceptors in limbs also play a big role in maintaining stability and balance.

ANIMAL BEHAVIOUR



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